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Constitutive model for rock fractures: Revisiting Barton's empirical model

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ABSTRACT

Barton's original model is modified in order to address the following limitations: the peak shear displacement is independent of normal stress and is zero for sawed fractures; for post-peak shear strength, Barton suggested a zero mobilized Joint Roughness Coefficient (*JRC*) after 100 times the peak shear displacement; Barton assumed zero dilation displacement at up to one-third of peak shear displacement. In this study, a database of results of direct shear tests available in the literature was collected and analyzed. An empirical equation is introduced to predict peak shear displacement, which considers the effect of normal stress and works for all types of rock discontinuities, even sawed fractures. The post-peak mobilized *JRC* can be obtained using a power-base equation, instead of employing Barton's table. This new empirical equation for post-peak mobilized *JRC* works for all ranges of displacements. The modified model can predict negative compressive dilatancy at small shear displacements.

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1. Introduction

In near-surface geotechnical works (for instance, dam foundations, power plants, underground caverns, and cuts), the mechanical behavior of the rock masses is influenced more by the fractures than by the intact rock. Therefore, analytical calculations and numerical simulations of the mechanical behavior of fractured rock masses require a constitutive law of rock fractures. However, the characteristics of intact rock are better known; for example, the Suggested Methods of the International Society of Rock Mechanics (Brown, 1981) define mathematically the Young's modulus for uniaxial compressive tests but leave out any calculations for the shear and normal stiffnesses of rock fractures (Muralha, 1990).

In the study of the behavior of a single rock fracture under different loading conditions, rock fractures are divided into two main categories: filled and unfilled fractures. The shear behavior of unfilled fractures is a function of the roughness and compressive strength of the fracture walls, while in the case of filled fractures, the physical and mineralogical properties of the material separating the fracture walls are of primary concern (Barton and Choubey, 1977). This paper investigates the constitutive models of unfilled rock fractures with dilatancy and surface degradation.

There are two main approaches to the quantitative description of the mechanical properties of rock fractures: (a) the theoretical approach, which adopts known theories (e.g. plasticity, contact theory, etc.) to simulate the observed behavior; (b) the empirical approach, in which physical data are analyzed to derive correlations between variables of influence and models are formulated according to observed behavior. Several empirical and theoretical constitutive models were developed by Patton (1966a), Ladanyi and Archambault (1969), Barton (1971), Goodman (1974, 1976), Barton and Choubey (1977), Hsu-Sun (1979), Bandis et al. (1981), Swan (1981), Heuze and Barbour (1981), Barton et al. (1985), Pande (1985), Kane and Drumm (1987), Plesha (1987), Desai and Fishman (1987, 1991), Amadei and Saeb (1990), Gens et al. (1990), Jing (1990), Jing et al. (1993), Qiu et al. (1993), Huang et al. (1993), Wibowo et al. (1993), Wibowo, (1994), Kana et al. (1996), Fox et al. (1998), Wang et al. (2003), Samadhiya et al. (2008), etc.

Many researchers have attempted to predict the shear strength of non-planar rock fractures based on their dilatant behavior. Jaeger (1959), Krsmanovic and Langof (1964), Lane and Heck (1964), Patton (1966b) and Byerlee (1975) are among those who first obtained curved relationships between the shear strength of the rock fractures and the normal stress.

Degradation of fracture asperities was investigated by Ladanyi and Archambault (1969), Plesha (1987), Zubelewicz et al. (1987), Hutson (1987), Saeb (1990), Hutson and Dowding (1990), Huang et al. (1993), Homand et al. (1999, 2001), and Lee et al. (2001).

The prediction of the dilatancy phenomenon of regular or irregular fractures subjected to direct shear loading has been addressed by numerous researchers such as Patton (1966a), Ladanyi and Archambault (1969), Jaeger (1971), Barton (1973, 1976), Leichnitz (1985), Saeb (1990), Homand et al. (1999, 2001), etc.

Among these models, Barton's empirical model (Barton, 1973) has widely been used because it is easy to apply and includes several

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important factors of fracture properties. In this paper, Section 2 recalls the essential aspects of Barton–Bandis empirical models. Section 3 addresses some of the limitations of Barton's model in predicting the peak shear displacement, post-peak shear strength, dilation, and surface degradation by using a database of results from direct shear tests available in the literature.

2. Barton–Bandis empirical models

2.1. The shear strength of rock fractures

Barton (1973) suggested the following empirical law of friction for the shear strength of rock fractures:

$$\tau = \sigma_n \cdot \tan\left(JRC \cdot \log_{10}\left(\frac{JCS}{\sigma_n}\right) + \phi_r\right) \quad (1)$$

For unweathered rock fractures, the residual friction angle, ϕ_r, is equal to base friction angle, ϕ_b, which can be obtained from shear tests on flat unweathered rock surfaces. The basic friction angle of the majority of unweathered rock surfaces ranges from 25° to 35°, at least at medium stress levels (Barton, 1971; Coulson, 1972; Hutchinson, 1972; Krsmanovic, 1967; Patton, 1966b; Ripley and Lee, 1962; Wallace et al., 1970). The residual friction angle of weathered rock fractures may be estimated based on the Schmidt hammer rebound number on dry unweathered sawed surfaces, r, and wet fracture surfaces, R, as follows (Barton and Choubey, 1977):

$$\phi_r = \left(\phi_r - 20^\circ\right) + 20 \cdot (r / R) \quad (2)$$

The joint roughness coefficient (JRC) represents a sliding scale of roughness varying from approximately 20 to 0, from the roughest to the smoothest rock surfaces. The joint compressive strength (JCS) at low stress levels is equal to the unconfined compression strength, σ_c, of the rock if the fracture is unweathered, but may reduce to approximately $\sigma_c/4$ for weathered fractures (Barton, 1971).

2.2. Peak shear displacement

The shear displacement, δ_{peak}, required to reach peak shear strength and defined in Fig. 1 determines the secant stiffness of fractures in shear. This is extremely important input data in finite element (Barton, 1972) and distinct element (Asadollahi, 2004) analyses.

Barton (1971) indicated that model tension fractures representing prototype fracture lengths from 225 cm up to 2925 cm required approximately 1% displacement to reach peak shear strength ($\delta_{peak} = 0.01L$). In addition, Barton and Choubey (1977) suggested 1% as a “rule-of-thumb” for discontinuities' peak shear strain, based on the overall mean obtained for 136 specimen ($\delta_{peak} = 0.0095L$). However, they pointed out that δ_{peak} will eventually reduce to less than $0.01L$ as fracture length increases to several meters. Barton and Bakhtar's (1983) survey of almost 300 shear test records revealed that peak shear displacement of lab-size fractures (224 tests) averaged at 1.28% of their corresponding lengths. On the other hand, 71 *in situ* tests gave an average of 0.72% of fracture lengths, thus yielding an overall average of 0.98%. Barton (1982), by reviewing a large number for shear tests reported in the literature (650 data points), found that the ratio δ_{peak}/L reduces gradually with increasing block length. Moreover, he proposed the following approximation to the mean trend of 170 data points:

$$\delta_{peak} = 0.004 \cdot L^{0.6} \quad (3)$$

The data points that Barton (1982) used to propose Eq. (3) included:

- 56 samples of clay-bearing discontinuities and rock joints ($5 \le L \le 15$ cm)
- 94 samples of clay-bearing discontinuities, rock joints, and rock models ($0.15 \le L \le 5$ m)
- 5 samples of clay-bearing discontinuities and model joints ($5 \le L \le 10$ m)
- 15 earthquake faults ($10 \le L \le 1000$ km).

Analysis of data published by Bandis et al. (1981) indicates that the ratio δ_{peak}/L is related to the JRC of the particular length of fractures tested, and that improved fit to the data is obtained with the following equation (Barton, 1982):

$$\frac{\delta_{peak}}{L} = \frac{1}{500} \cdot \left(\frac{JRC}{L}\right)^{0.33} \quad (4)$$

where L is length of fracture sample (in meters).

2.3. Degradation of fracture asperity

The degradation of fracture asperity can be conceptualized as the variation of asperity angle, which would be evaluated by the secant or tangential slope of dilation curves. Barton (1982) showed that the shear stress–shear displacement curve may be obtained by using shear displacement dependent JRC values in Eq. (1). These JRC values are called $JRC_{mobilized}$ and are given in Table 1. On the other hand, by starting from a measured shear stress–shear displacement curve, $\tau(\delta_h)$, the magnitude of $JRC_{mobilized}$ can be calculated as follows:

$$JRC_{mobilized}(\delta_h) = \frac{\arctan(\tau(\delta_h) / \sigma_n) - \phi_r}{\log(JCS / \sigma_n)} \quad (5)$$

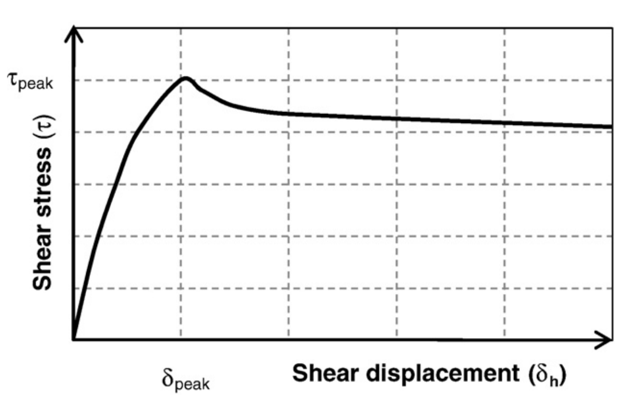

Fig. 1. Typical shear stress–displacement curve.

Table 1
Recommended model for shear stress–displacement (Barton, 1982).

Non-planar fractures		Planar fractures ($JRC_{peak} \le 5$)	
$\frac{\delta_h}{\delta_{peak}}$	$\frac{JRC_{mobilized}}{JRC_{peak}}$	$\frac{\delta_h}{\delta_{peak}}$	$\frac{JRC_{mobilized}}{JRC_{peak}}$
0	$-\phi_r/i$[a]	0	$-\phi_r/i$[a]
0.3	0	0.3	0
0.6	0.75	0.6	0.75
1.0	1.0	1.0	0.95
2.0	0.85	2.0	1.0
4.0	0.70	4.0	0.9
10.0	0.50	10.0	0.7
25.0	0.40	25.0	0.5
100	0	100	0

[a] $i = JRC_{peak} \cdot \log(JCS/\sigma_n)$.

The ratio $JRC_{\mathrm{mobilized}}/JRC_{\mathrm{peak}}$ can be estimated from the ratio $\delta_h/\delta_{\mathrm{peak}}$ using the values given in Table 1.

2.4. Dilatancy

The peak secant dilation angle (also called initial dilation angle), $d_{s,\mathrm{peak}}$, and the peak tangent dilation angle, $d_{t,\mathrm{peak}}$, are depicted in Fig. 2 and defined as follows:

$$d_{s,\mathrm{peak}} = \arctan\left(\frac{\delta_{v,\mathrm{peak}}}{\delta_{\mathrm{peak}}}\right) \tag{6}$$

$$d_{t,\mathrm{peak}} = \left(\frac{\partial\delta_v}{\partial\delta_h}\right)_{\mathrm{at}\ \delta_h=\delta_{\mathrm{peak}}} \tag{7}$$

Barton and Choubey (1977) showed that the peak secant dilation angle is about one-third of the peak tangent dilation angle and both can occasionally be negative or zero. In addition, they suggested the following relation for the peak secant and tangent dilation angles (Barton and Choubey, 1977):

$$d_{t,\mathrm{peak}} = (1/M)\cdot JRC\cdot \log_{10}(JCS/\sigma_n) \tag{8}$$

$$d_{s,\mathrm{peak}} = (1/3)\cdot JRC\cdot \log_{10}(JCS/\sigma_n), \tag{9}$$

where M is a damage coefficient, that takes values of 1 or 2 for shearing under low or high normal stress, respectively (Olsson and Barton, 2001), or can be obtained from the following relationship (Barton and Choubey, 1977):

$$M = \frac{JRC}{12\cdot \log_{10}(JCS/\sigma_n)} + 0.70 \tag{10}$$

Barton (1982) indicated that dilation will begin when $JRC_{\mathrm{mobilized}} = 0$ and mobilized dilation angle can be obtained from the following relationship:

$$d_{t,\mathrm{mobilized}} = (1/M)\cdot JRC_{\mathrm{mobilized}}\cdot \log_{10}(JCS/\sigma_n) \tag{11}$$

3. Modified Barton's model

Barton's failure criterion has been widely used to estimate the shear strength of rock factures because it is easy to apply and includes several important fracture properties that can be easily measured or estimated (Grasselli and Egger, 2003). In addition, Grasselli and Egger (2003) stated that researchers studying the contribution of morphology to the shear strength should relate their model to the JRC criterion proposed by Barton in the 1970s (Barton, 1973; Barton and Choubey, 1977), and adopted as a reference by the International Society of Rock Mechanics (ISRM, 1978).

Although, Barton's failure criterion predicts the peak shear strength of rock fractures with acceptable precision, it shows the following limitations in estimating the peak shear displacement, post-peak shear strength, dilation, and surface degradation:

- The peak shear displacement is independent of normal stress and is equal to zero for sawed fractures (Bandis et al., 1983); this is not consistent with experimental observations. For example, peak shear displacements between 0.05 and 2.71 mm were reported in the literature (Desai and Fishman, 1991; Huang et al., 2002; Yoshinaka and Yamabe, 1986).
- Barton suggested that the mobilized JRC should be equal to zero when the shear displacement is greater than 100 times the peak shear displacement. This seems to be just an approximation for the end of the shear stress–displacement curve because there are few experimental results containing post-peak shear strength of rock fractures up to about 100 times the peak shear displacement. Moreover, even after this amount of displacement, the fracture surface is not the same as a sawed fracture ($JRC_{\mathrm{mobilized}} = 0$).
- Barton divided the problem of post-peak shear strength into two categories (Table 1): non-planar ($JRC>5$) and planar ($JRC\leq 5$) fractures (Barton, 1982). There is an inconsistency for the case of planar fracture. Although JRC_{peak} should be mobilized at the peak shear displacement (δ_{peak}), Table 1 gives the following magnitudes for planar fractures:
  - ○ $\delta_h = \delta_{\mathrm{peak}}$: $JRC_{\mathrm{mobilized}} = 0.95 JRC_{\mathrm{peak}}$
  - ○ $\delta_h = 2\delta_{\mathrm{peak}}$: $JRC_{\mathrm{mobilized}} = JRC_{\mathrm{peak}}$
- Barton assumed zero dilation (displacement) up to one-third of peak shear displacement and disregarded negative dilatancy. However, many experimental studies performed on rock fractures showed that there is a negative dilation at small shear displacements (Homand et al., 2001; Huang et al., 2002; Jacobsson, 2004; Jacobsson, 2005a; Jacobsson, 2005b; Jacobsson, 2006a; Jacobsson, 2006b; Jacobsson and Flansbjer, 2005a; Jacobsson and Flansbjer, 2005b; Jacobsson and Flansbjer, 2006a; Jacobsson and Flansbjer, 2006b; Olsson and Barton, 2001).
- On the other hand, Barton (1982) proposed Eq. (11) for the tangent dilation angle and Table 1 for mobilized JRC. Based on Table 1, $JRC_{\mathrm{mobilized}}$ is negative up to $\delta/\delta_{\mathrm{peak}} = 0.3$. Therefore, the tangent dilation angle should be negative up to $\delta/\delta_{\mathrm{peak}} = 0.3$. In addition, $JRC_{\mathrm{mobilized}}$ is zero at $\delta/\delta_{\mathrm{peak}} = 0.3$ and then it has a positive value. As a result, dilation should be negative up to $\delta/\delta_{\mathrm{peak}} = 0.3$ and then positive. Thus, the minimum dilation displacement, achieved at $\delta/\delta_{\mathrm{peak}} = 0.3$ not at the point where the fracture starts to dilate.

In this study, the original Barton model is modified to remove these limitations. There are two main approaches to the quantitative description of the mechanical properties of rock fractures: the theoretical approach, and the empirical approach (Bandis, 1990). In this research, the empirical approach was used, which is consistent with Barton's empirical model. Barton's failure criterion for peak shear strength of rock fracture, Eq. (1), was adopted. A database was built by collecting the results of 362 direct shear tests available in the literature: Monotonic Direct Shear Tests, called MDST (see Appendix A for the database). The ability of Barton's model to predict peak shear displacement, dilation, and post-peak shear strength was investigated and modifications are proposed to improve it.

3.1. Peak shear displacement

The purpose of this section is to find an empirical relationship that can also represent the behavior of smooth fractures ($JRC = 0$) for peak shear displacement, δ_{peak} considering the effect of normal stress. The value of δ_{peak} may be affected by the length of the block (L), JRC, JCS, and the normal stress (σ_n). The peak shear displacement and the block length have length dimension (L). In addition, JCS and

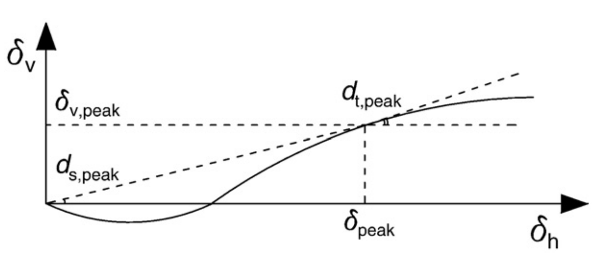

Fig. 2. Typical shear displacement–dilation curve.

normal stress have stress dimension (FL^{-2}). *JRC* is dimensionless. Dimensional analysis (Szirtes, 1997) led to the following dimensionless parameters:

$$\pi_1 = \frac{\delta_{peak}}{L}, \pi_2 = \frac{L}{L_0}, \pi_3 = JRC, \pi_4 = \frac{\sigma_n}{JCS},$$

where L_0 is length of lab specimen (0.1 m).

Correlation analyses were performed to find π_1 as a function of π_2, π_3, π_4:

$$\pi_1 = f(\pi_2, \pi_3, \pi_4) \tag{12}$$

The MDST database contained 362 direct shear test records. Cases that have *JRC* values between 2 and 20 were selected for this part of analyses (317 data points).

In order to perform a reliable correlation analysis, all variables should have reasonable distributions. The block lengths in the MDST database ranged from 0.049 to 3 m with the distribution depicted in Fig. 3. Since the MDST database contained the results of laboratory direct shear tests, the majority of block sizes is centered around 0.1 m. Therefore, the database may under-represents long fractures. However, the following paragraphs demonstrate that the MDST database is adequate from this point of view.

Barton (1982) found an approximation to the mean trend of 170 data, Eq. (3), where block length and earthquake faults ranged from about 50 mm to 1000 km. Non-linear regression analysis performed on the MDST database to correlate peak shear displacement and length of the sample gave the following relationship:

$$\delta_{peak} = 0.0032 \cdot L^{0.61} \tag{13}$$

As can be seen in Fig. 4, the predicted values from Eqs. (3) and (13) are close to each other and their relative difference is less than 15%. It is almost impossible to collect all required information (such as *JRC*, *JCS*, normal stress, peak shear stress, and base friction angle) from sheared large blocks. Even Barton could only collect the values of peak shear displacement and length of the block for large blocks. Therefore, the MDST database is adequate from this point of view.

The magnitude of *JRC* in the MDST ranged between 2 and 20 as depicted in Fig. 5(a). In addition, σ_n/JCS ranged between 0.001 and 0.6 with the distribution illustrated in Fig. 5(b).

A power-base relationship has been adopted by Barton to relate peak shear displacement with block's length and *JRC*, Eq. (4). Moreover, the power function is a convenient form to use in calculations. Consequently, it was first assumed that function *f* in Eq. (12) is a power function. With coefficient of correlation $R^2 = 0.42$ and standard error of estimate equal to 0.65, the following equation for peak shear displacement of the rock fracture was derived from correlation analysis (*L* in m):

$$\delta_{peak} = 0.02 \frac{L^{0.51}}{JRC^{0.37}} \left(\frac{\sigma_n}{JCS}\right)^{0.32} \tag{14}$$

There is a major difference between Barton's empirical relationship, Eq. (4), and what is obtained here by correlation analysis in Eq. (14): although Barton (by correlation analysis of 130 data points) found that peak shear displacement increases with *JRC*, the opposite is found here (by correlation analysis of 317 data points). Indeed, the single Patton's (1966a,b) model and analytical calculations described in the Appendix B show that the peak shear displacement should decrease with increasing *JRC*.

Although the predicted peak shear displacement using Eq. (14) changes with normal stress and decreases with increasing *JRC*, Eq. (14) still breaks down in the case of smooth fractures ($JRC \to 0$). Therefore, the regression analysis was revised and the following empirical relation was derived (see Appendix C for derivation):

$$\delta_{peak} = 0.0077 L^{0.45} \left(\frac{\sigma_n}{JCS}\right)^{0.34} \cos\left(JRC \log\left(\frac{JCS}{\sigma_n}\right)\right) \tag{15}$$

Table 2 compares the ratio of predicted to the measured peak shear displacement, $\left(\frac{\delta_{predicted}}{\delta_{measured}}\right)$, obtained using Eqs. (4), (14), and (15) for the 317 cases of MDST database with *JRC* between 2 and 20. Knowing the fact that the ideal value for $\left(\frac{\delta_{predicted}}{\delta_{measured}}\right)$ is one, based on the following reasons, it can be concluded that Eq. (15) works the best in predicting peak shear displacement of rock fractures:

- Although Eq. (14) has the closet value of $\left(\frac{\delta_{predicted}}{\delta_{measured}}\right)_{Min}$ to one among the other options, the values of $\left(\frac{\delta_{predicted}}{\delta_{measured}}\right)_{Average}$, $\left(\frac{\delta_{predicted}}{\delta_{measured}}\right)_{Max}$ obtained using Eq. (15) are closer to one compared to those determined using Eqs. (4) and (14).
- Eq. (15) has the minimum sum of the square of errors, $\sum\left(1-\frac{\delta_{predicted}}{\delta_{measured}}\right)^2$, among the other options. The smaller sum of the square of errors is, the better the empirical equation is.
- The correlation factor, which is defined as the ratio of standard deviation to the average, $\left(\frac{\delta_{predicted}}{\delta_{measured}}\right)_{STD} / \left(\frac{\delta_{predicted}}{\delta_{measured}}\right)_{Average}$, is also the minimum in the case of Eq. (15) (0.68 compared to 0.99 and 0.69 from Barton's equation and Eq. (14), respectively). The smaller the correlation factor is, the better the empirical equation is.

In addition, for the above mentioned 317 data points, Fig. 6 illustrates predicted peak shear displacement, $\delta_{predicted}$, using Barton's

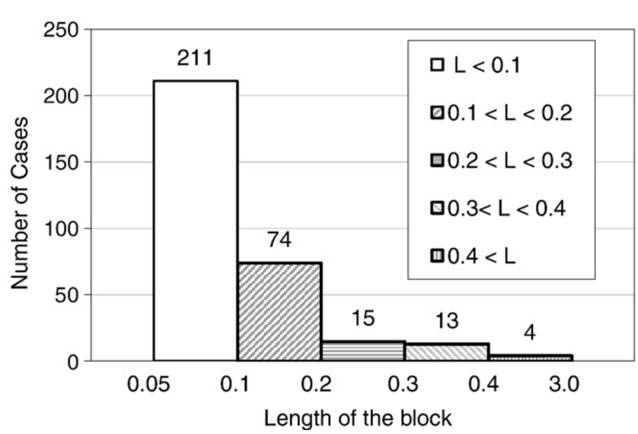

Fig. 3. Distribution of length of the block in 317 data points of the database used in correlation analysis of peak shear displacement.

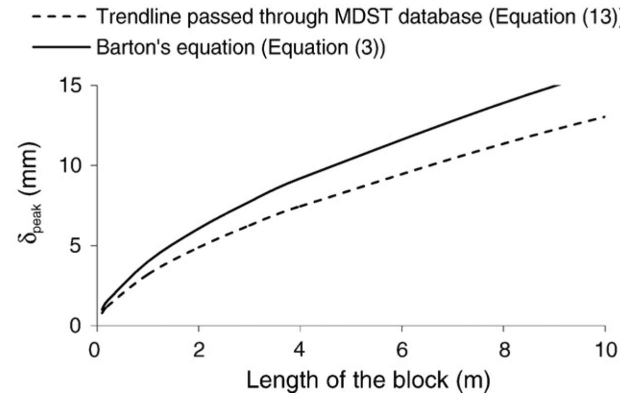

Fig. 4. Comparison between Barton's equation correlating peak shear displacement and length of the block (Eq. (3)) and trend line passed through our database (Eq. (13)).

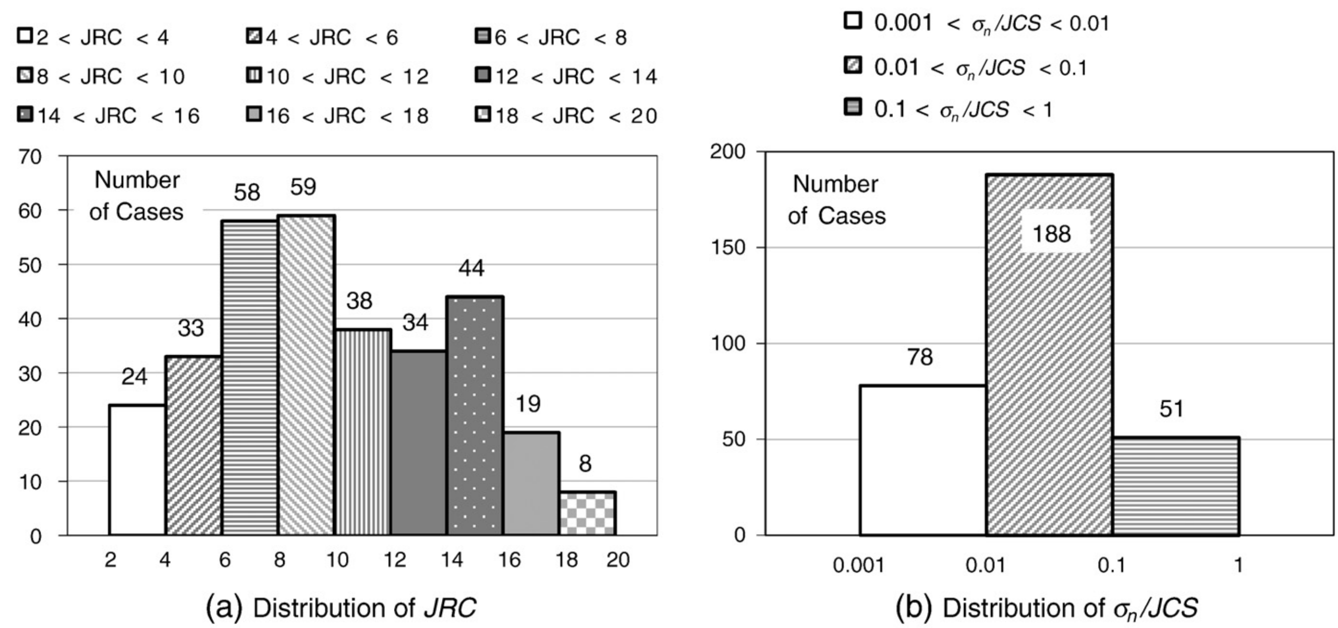

Fig. 5. Distribution of *JRC* and σ_n/JCS in 317 data points of the database used in correlation analysis of peak shear displacement.

equation and Eq. (15) versus the measured peak shear displacement, $\delta_{measured}$. The distribution of $\delta_{predicted}$ versus $\delta_{measured}$ for the case of Eq. (15) is closer to the ideal line $\delta_{predicted} = \delta_{measured}$ compared to those obtained employing Barton's equation.

Furthermore, using Eq. (15) to predict the peak shear displacement of rock fractures has advantages over Eqs. (4) and (14), because it is the only one that can be used for all types of rock fractures including sawed, smooth, and rough. Fig. 7 compares experimental peak shear displacement for sand blasted and sawed fractures ($JRC = 0$) with the values predicted employing Eq. (15), as the suggested empirical equation of this study. While Eq. (4) predicts zero peak shear displacement for sawed fractures and Eq. (14) tends to infinity, Eq. (15) generally yields a good estimation, as it can be seen in Fig. 7. Some over/under estimation occurs in isolated cases.

Fig. 8 demonstrates the ability of Eq. (15) to consider the effect of normal stress on the peak shear displacement. Also shown in Fig. 8 is Wibowo's linear correlation ($\delta_{peak} = a + b \cdot \sigma_n$). Although, Wibowo's approximation performs the best in these figures, constants *a* and *b* in Wibowo's model have to be determined experimentally for each fracture, whereas no additional parameter has to be determined in Eq. (15).

In conclusion, the empirical Eq. (15) is proposed to predict the peak shear displacement of rock fractures. This equation considers the effect of normal stress on the peak shear displacement, while Barton's equation does not. In addition, Eq. (15) can be used for all types of rock fractures, including sawed, smooth, and rough, while Barton's equation predicts an incorrect value of zero for the peak shear displacement of sawed fractures. Finally, predicted peak shear

Table 2
Comparing Barton's empirical equation (Eq. (4)) with Eqs. (14) and (15) in predicting the peak shear displacement of rock fractures.

Parameter	Barton's equation (Eq. (4))	Eq. (14)	Eq. (15)
$\left(\frac{\delta_{predicted}}{\delta_{measured}}\right)_{Average} \pm STD$	1.64 ± 1.63	1.26 ± 0.87	1.11 ± 0.76
$\left(\frac{\delta_{predicted}}{\delta_{measured}}\right)_{max}$	13.70	6.36	5.59
$\left(\frac{\delta_{predicted}}{\delta_{measured}}\right)_{min}$	0.12	0.23	0.19
$\sum\left(1-\frac{\delta_{predicted}}{\delta_{measured}}\right)^2$	980.22	259.70	188.47

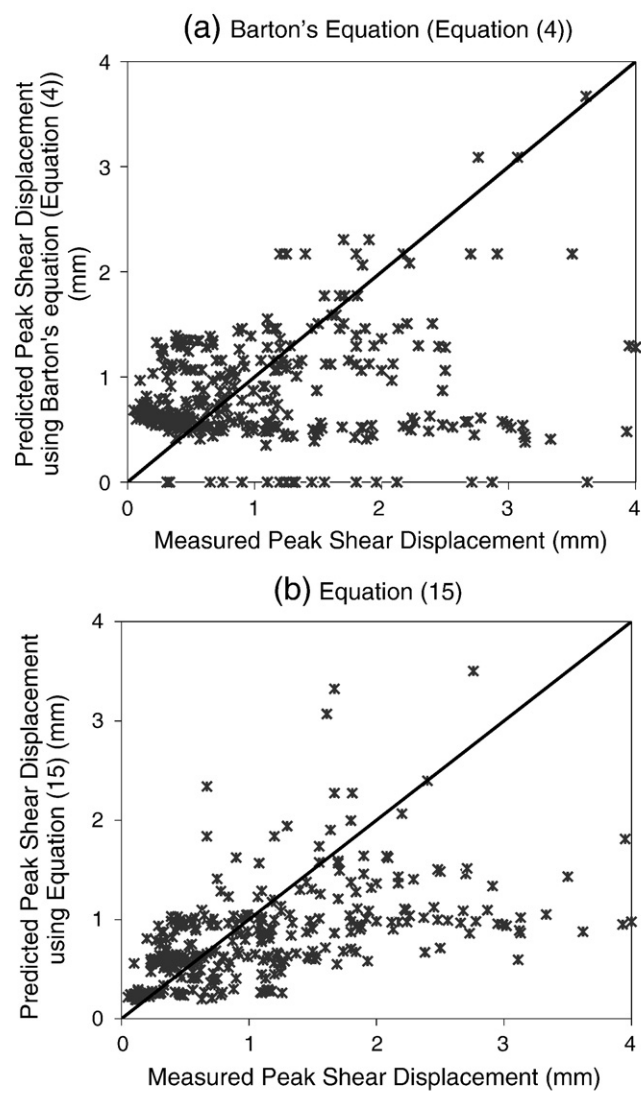

Fig. 6. Predicted versus the measured peak shear displacement.

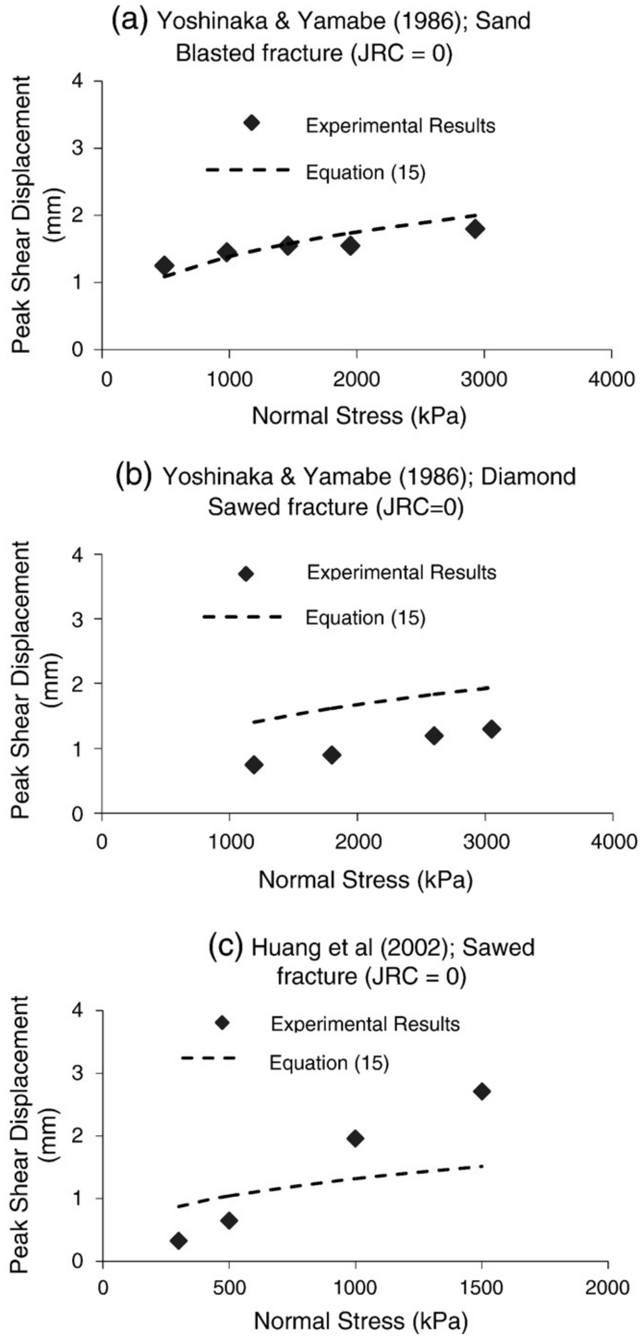

Fig. 7. Comparing measured peak shear displacement with their predicted values using Eq. (15) for the case of sawed fractures (*JRC* = 0).

displacement employing Eq. (15) decreases as *JRC* increases. However, the predicated value of peak shear displacement using Barton's equation increases with *JRC*.

3.2. Post-peak shear strength

The MDST database was analyzed in order to find an empirical relationship between the mobilized *JRC* and post-peak shear displacement. Initially, it was assumed that the ratio of $JRC_{\text{mobilized}}/JRC_{\text{peak}}$ is a function of not only $\delta_h/\delta_{\text{peak}}$, but also normal stress, *JCS*, and JRC_{peak}.

It should be noted that most of the direct shear tests performed for research or professional purposes are conducted up to a shear displacement of not more than 15 mm. Even according to ASTM D5607-02, the displacement devices used to measure shear displacement in direct shear tests should accommodate a displacement of 13 mm. For a lab-size specimen of 10 cm, the peak shear displacement is about 1 mm (using the "rule-of-thumb" suggested by Barton (1971)) and shear displacement of 15 mm is approximately 15 times the peak shear displacement. Consequently, any database used for correlation analysis suffers from lack of information for post-peak shear strength when shear displacements are greater than $15\delta_{\text{peak}}$.

The MDST database contains 255 direct shear test records for which the magnitude of post-peak shear strength are known at 1 to 4 different points; this gives a total number of 758 data points with $\delta_h/\delta_{\text{peak}}$ ranging between 1 and 25. The 758 data points have a distribution illustrated in Fig. 9(a), which shows that most of the data points have $\delta_h/\delta_{\text{peak}}$ between 1 and 15 (730 out of 758).

Fig. 9(a) shows that any regression analysis performed on the MDST database is reliable up to $\delta_h/\delta_{\text{peak}} = 10$. Clearly, for $\delta_h/\delta_{\text{peak}} > 10$, the obtained relationship from a correlation analysis is almost an extrapolation of the approximation through the points with smaller amount of displacements. Thus, for large shear displacements, the correlation analysis of MDST suffers from exactly the same weakness as Barton's model.

By performing a dimensional analysis (Szirtes, 1997), the following dimensionless parameters were found:

$$\pi_3 = JRC_{\text{peak}},\ \pi_4 = \frac{\sigma_n}{JCS},\ \pi_5 = \frac{JRC_{\text{mobilized}}}{JRC_{\text{peak}}},\ \pi_6 = \frac{\delta_h}{\delta_{\text{peak}}}$$

Correlation analyses were performed to find π_5 as a function of π_3, π_4, and π_6:

$$\pi_5 = f(\pi_3, \pi_4, \pi_6) \tag{16a}$$

The magnitude of JRC_{peak} in 758 data points of the MDST database ranges from 0 to 20 with a normal distribution depicted in Fig. 9(b). In addition, Fig. 5(a) shows $\frac{\sigma_n}{JCS}$ ranged between 0.001 and 0.6 with an acceptable distribution. Initial correlation analysis showed that there is no correlation between π_5 and both π_3 and π_4. Therefore, Eq. (16a) can be written as follows:

$$\pi_5 = f(\pi_6) \tag{16b}$$

Since at peak shear displacement ($\delta_h = \delta_{\text{peak}}$), the mobilized *JRC* should be equal to JRC_{peak}, Eq. (16b) should satisfy the condition $\pi_5 = f(1) = 1$.

The correlation analysis revealed that a power function fits experimental results the best. Thus, the following empirical equation with $R^2$of 0.52 and standard error of estimate of 0.58 is proposed to obtain $JRC_{\text{mobilized}}/JRC_{\text{peak}}$ from $\delta/\delta_{\text{peak}}$:

$$\frac{JRC_{\text{mobilized}}}{JRC_{\text{peak}}} = \left(\frac{\delta_{\text{peak}}}{\delta_h}\right)^{0.381} \tag{17}$$

Table 3 compares the predicted versus measured values obtained by using Eq. (27) against Table 1 for 758 data points of the MDST database. Based on the following reasons, it can be concluded from Table 3 that Eq. (17) works better than Table 1 in predicting $JRC_{\text{mobilized}}$:

- Although Table 3 has the closer value of $\left(\frac{(JRC_{\text{mobilized}})_{\text{predicted}}}{(JRC_{\text{mobilized}})_{\text{measured}}}\right)_{\text{Min}}$ to one, the values of $\left(\frac{(JRC_{\text{mobilized}})_{\text{predicted}}}{(JRC_{\text{mobilized}})_{\text{measured}}}\right)_{\text{Average}}$, $\left(\frac{(JRC_{\text{mobilized}})_{\text{predicted}}}{(JRC_{\text{mobilized}})_{\text{measured}}}\right)_{\text{Max}}$ are closer to one compared to those obtained from Table 1.
- Eq. (17) has the smaller sum of the square of errors, $\sum\left(1-\frac{\delta_{\text{predicted}}}{\delta_{\text{measured}}}\right)^2$than Table 1. The smaller sum of the square of errors is, the better the empirical equation is.

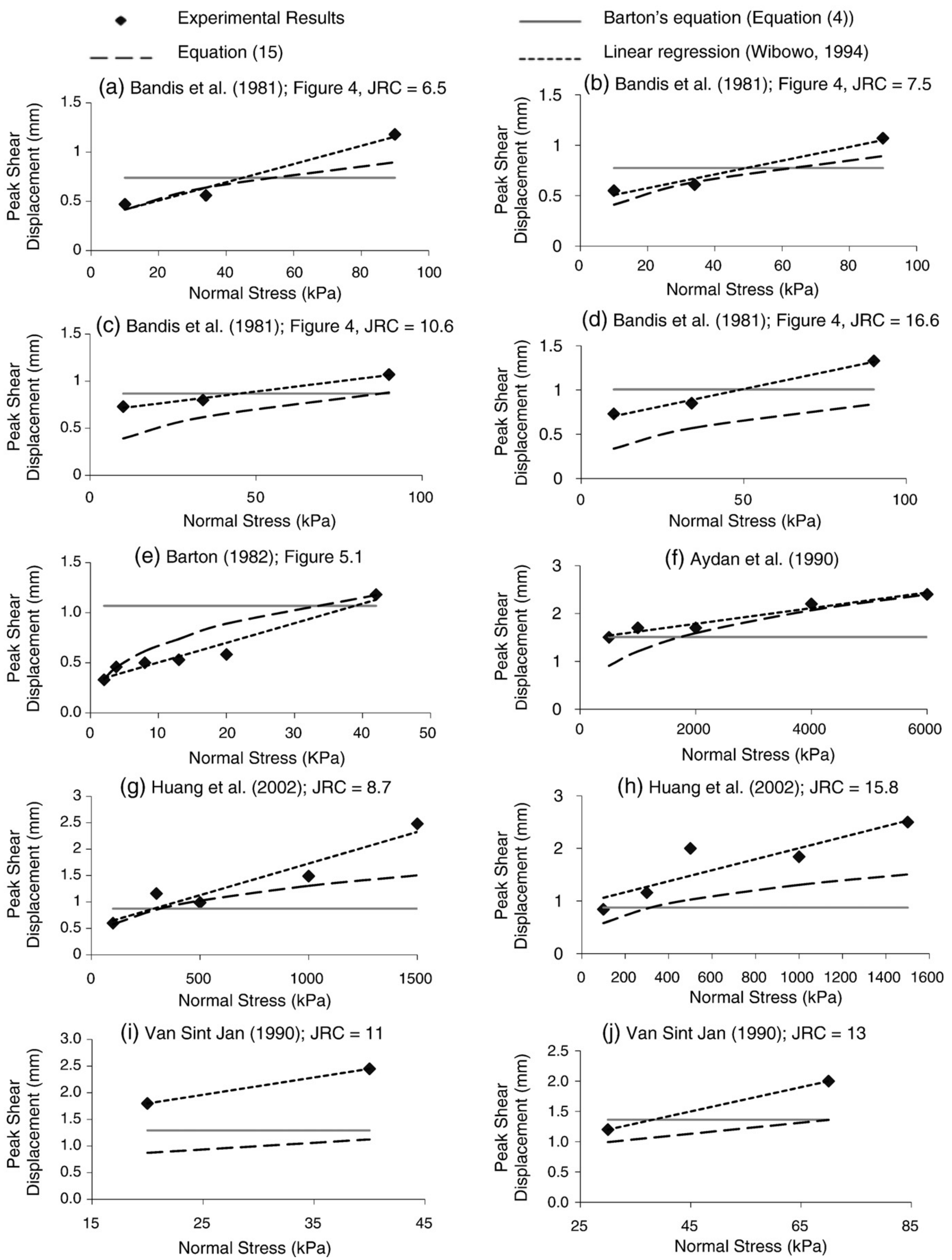

Fig. 8. Performance of Eq. (15) in considering the effect of normal stress on the peak shear displacement as compared to Barton's equation (Eq. (4)).

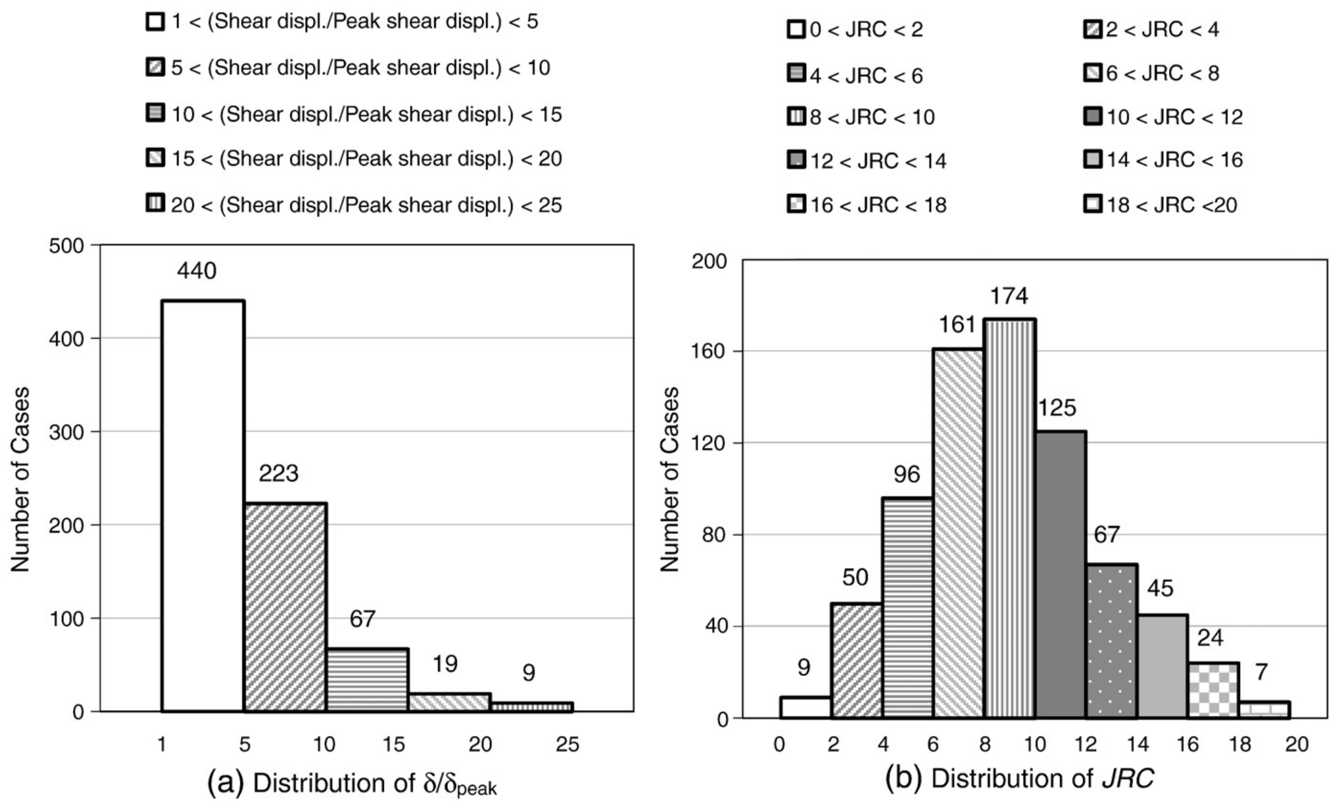

Fig. 9. Distribution of δ/δ_{peak} and *JRC* for 758 data points of MDST database used in correlation analysis of post-peak shear strength.

– The correlation factor, which is defined as the ratio of standard deviation to the average, $\dfrac{\left(\frac{(JRC_{\mathrm{mobilized}})_{\mathrm{predicted}}}{(JRC_{\mathrm{mobilized}})_{\mathrm{measured}}}\right)_{\mathrm{STD}}}{\left(\frac{(JRC_{\mathrm{mobilized}})_{\mathrm{predicted}}}{(JRC_{\mathrm{mobilized}})_{\mathrm{measured}}}\right)_{\mathrm{Average}}}$, is also the minimum in the case of Eq. (17) (0.77 compared to 0.79). The smaller is the correlation factor, the better works the empirical equation.

For a comparison, Fig. 10 depicts the proposed model, Eq. (17), and Barton's model (Table 1) for δ_h/δ_{peak} ranging between 1 and 100.

In conclusion, Eq. (17) is proposed to predict the mobilized *JRC* after peak shear displacement. In addition to the fact that it works better than Table 1 in the MDST database, it has a smoother curve compared to the linear interpolation of the values given in Table 1 and is easier to implement numerically. Furthermore, Eq. (17) is independent of *JRC* and does not have the same problem as the above mentioned inconsistency of Table 1 for the case of planar fractures (Section 3). However, both Table 1 and Eq. (17) suffer from the same problem: lack of information of rock fracture shear strength at shear displacement greater than 10 times the peak shear displacement. As a result, the predicted mobilized *JRC* at high shear displacements ($\delta_h/\delta_{peak}>10$) using either Table 1 or Eq. (17) are just an extrapolation of the relationships obtained by correlation analyses of data points available at smaller displacements. However, in this regard Eq. (17) has the following advantage over Table 1; at $\delta_h/\delta_{peak}=100$, Table 1 suggested $JRC_{\mathrm{mobilized}}=0$, while Eq. (17) proposed $JRC_{\mathrm{mobilized}}=0.17JRC_{\mathrm{peak}}$, which is more realistic, because, even after this large amount of displacement, one cannot expect a rough fracture to behave the same as a sawed fracture.

Up to about 50 times the peak shear displacement, the predicted $JRC_{\mathrm{mobilized}}$ using Eq. (17) is smaller than that obtained employing Table 1. Before reaching 50 times the peak shear displacement of fractures, using Eq. (17) instead of Table 1 for post-peak shear displacement is conservative.

Table 3
Comparison between Barton's model proposed in tabular format (Table 1) and Eq. (17) in predicting the ratio of $JRC_{\mathrm{mobilized}}$ from real δ/δ_{peak}.

Parameter	Barton's model (Table 1)	Eq. (17)
$\left(\frac{(JRC_{\mathrm{mobilized}})_{\mathrm{predicted}}}{(JRC_{\mathrm{mobilized}})_{\mathrm{measured}}}\right)_{\mathrm{Average}} \pm \mathrm{STD}$	1.39 ± 1.10	1.19 ± 0.92
$\left(\frac{(JRC_{\mathrm{mobilized}})_{\mathrm{predicted}}}{(JRC_{\mathrm{mobilized}})_{\mathrm{measured}}}\right)_{\max}$	11.97	9.67
$\left(\frac{(JRC_{\mathrm{mobilized}})_{\mathrm{predicted}}}{(JRC_{\mathrm{mobilized}})_{\mathrm{measured}}}\right)_{\min}$	0.56	0.43
$\sum\left(1-\frac{(JRC_{\mathrm{mobilized}})_{\mathrm{predicted}}}{(JRC_{\mathrm{mobilized}})_{\mathrm{measured}}}\right)^2$	37.83	32.35

3.3. Dilatancy

In this section, the dilatant behavior of rock fractures is investigated. The goal is to find the dilation (displacement) at each shear displacement. The problem is divided into two parts: pre-peak and post-peak dilatancy, which means dilatancy before and after peak shear displacement, respectively.

3.3.1. Pre-peak dilatancy

The MDST database contains the results of 242 direct shear tests for which at least dilation at the peak shear displacement is available. Based on the shape of the vertical displacement versus the shear displacement curve and how much information is available for each test, the results were divided into 4 different categories (Table 4).

Barton and Choubey (1977) used the peak secant dilation angle, also called initial dilation angle, and the peak tangent dilation angle. Based on their experimental results, they found that the peak secant dilation angle is about one-third of the peak tangent dilation angle. They proposed Eqs. (8) and (9) for peak tangent and secant dilation

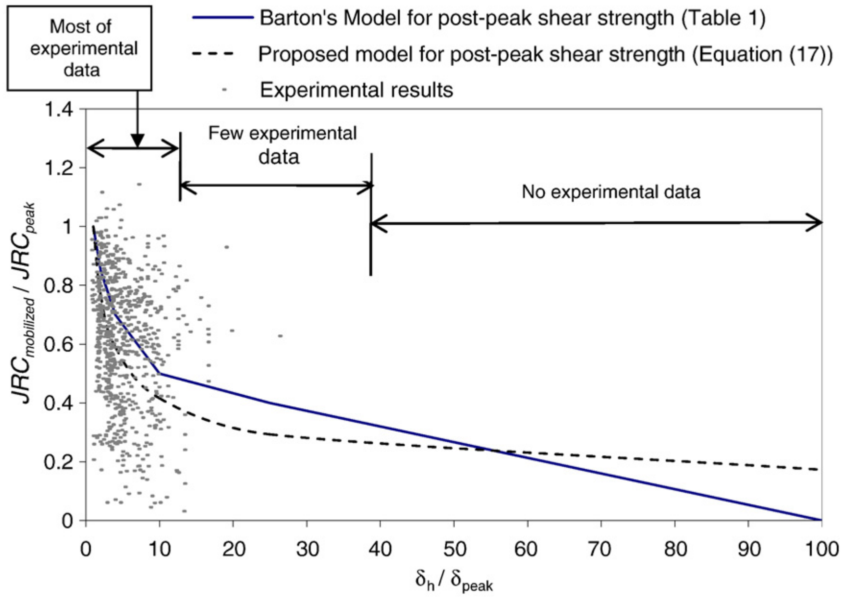

Fig. 10. Comparison between proposed model for post-peak shear strength (Eq. (17)) and Barton's model (Table 1).

angles, respectively. There are two ways to predict the peak secant dilation angle:

– Option 1: estimate the peak secant dilation angle using Eq. (9).
– Option 2: use Eq. (8) to estimate the peak tangent dilation angle, employ Eq. (10) to predict damage coefficient, M, and estimate the peak secant dilation angle to be one-third of the peak tangent dilation angle:

$$d_{s,\text{peak}} = 1/3\left(\frac{1}{M}JRC\cdot \log\left(\frac{JCS}{\sigma_n}\right)\right) \tag{18}$$

In order to find which option works better, all 204 cases (Categories 1, 2, and 4 of the MDST database in Table 4) were considered. For each case, the peak secant dilation angle, $d_{s,\text{peak}}$, was estimated using the above mentioned two options. Then, dilation displacement at peak shear displacement, $\delta_{v,\text{peak}}$, was calculated using the following equation with the measured value of δ_{peak}:

$$\delta_{v,\text{peak}} = \delta_{\text{peak}}\cdot \tan\left(d_{s,\text{peak}}\right) \tag{19}$$

The results given in Table 5 show that Option 2 is better than Option 1.

Next, the goal was to find at which shear displacement(s), dilation displacement is zero. Category 2 of the MDST database is the only part of this database that includes all dilatancy information (maximum negative dilation, dilation at peak shear displacement, and shear displacement at which dilation is zero). Analysis of Category 2 of the MDST database shows that, on average, zero dilation occurs at $\delta/\delta_{\text{peak}} = 0.5$.

Finally, the goal was to find an equation with which dilation displacement can be obtained at each shear displacement. The dimensionless forms of displacement are defined as $\frac{\delta_h}{\delta_{\text{peak}}}$ and $\frac{\delta_v}{\delta_{\text{peak}}}$ where δ_v is normal displacement (dilation displacement). The equation should satisfy the following conditions:

1) At $\frac{\delta_h}{\delta_{\text{peak}}} = 0$: $\frac{\delta_v}{\delta_{\text{peak}}} = 0$
2) At $\frac{\delta_h}{\delta_{\text{peak}}} = 0.5$: $\frac{\delta_v}{\delta_{\text{peak}}} = 0$
3) At $\frac{\delta_h}{\delta_{\text{peak}}} = 1$: $\frac{\delta_v}{\delta_{\text{peak}}} = \frac{\delta_{v,\text{peak}}}{\delta_{\text{peak}}}$

Therefore, a quadratic equation with zero intercept perfectly satisfies all three conditions:

$$\frac{\delta_v}{\delta_{\text{peak}}} = \frac{\delta_{v,\text{peak}}}{\delta_{\text{peak}}}\left[2\cdot\left(\frac{\delta_h}{\delta_{\text{peak}}}\right)^2 - \left(\frac{\delta_h}{\delta_{\text{peak}}}\right)\right] \tag{20}$$

Table 4
Different categories of the MDST database Based on the shape of the vertical displacement versus the shear displacement curve and available information for each test.

Categories	Number of test records	Available data	Issues
Category 1	96	Dilation at peak shear displacement Shear displacement at which the fracture started to dilate	No negative dilation was depicted. This category contains the results of 96 tests, 34 cases of which come from Barton (1982), Barton et al. (1985), and Bandis et al. (1981). While they did not consider the negative dilation in their model, there is a possibility that they had eliminated the negative part in presenting their experimental work.
Category 2	91	Maximum negative dilation Dilation at peak shear displacement Shear displacement at which dilation is zero	Fractures initially showed negative dilation followed by positive dilation.
Category 3	38		Dilation was negative at all points. 35 out of 38 cases of this category have experienced shearing under different normal stresses. Thus, there is considerable uncertainty regarding these data including mismatching results. Therefore, this category was not considered in correlation analyses.
Category 4	17	Dilation at peak shear displacement	

Table 5
Comparison of two available options to predict the peak secant dilation angle at peak shear displacement.

Parameter	Option 1	Option 2
$\left(\frac{(\delta_{v,peak})_{predicted}}{(\delta_{v,peak})_{measured}}\right)_{Average} \pm STD$	1.52 ± 2.14	1.21 ± 1.41
$\left(\frac{(\delta_{v,peak})_{predicted}}{(\delta_{v,peak})_{measured}}\right)_{Max}$	15.57	9.49
$\left(\frac{(\delta_{v,peak})_{predicted}}{(\delta_{v,peak})_{measured}}\right)_{Min}$	0.09	0.10
$\sum\left(1-\frac{(\delta_{v,peak})_{predicted}}{(\delta_{v,peak})_{measured}}\right)^2$	706.18	325.54

Based on the proposed model for pre-peak dilation displacement, the mobilized tangent dilation angle of rock fractures at each point can be obtained as follows:

$$d_{t,mobilized} = \arctan\left(\frac{\partial\left(\frac{\delta_v}{\delta_{peak}}\right)}{\partial\left(\frac{\delta_h}{\delta_{peak}}\right)}\right) = \arctan\left(\frac{\delta_{v,peak}}{\delta_{peak}}\left[4\cdot\left(\frac{\delta_h}{\delta_{peak}}\right)-1\right]\right) \tag{21}$$

Therefore, the peak tangent dilation angle can be obtained as follows:

$$d_{t,peak} = \arctan\left(3\frac{\delta_{v,peak}}{\delta_{peak}}\right) \tag{22}$$

Substituting Eq. (19) in Eq. (22), we have:

$$d_{t,peak} = \arctan\left(3\tan\left(d_{s,peak}\right)\right) \tag{23}$$

Recall that the shear strength of a rock fracture at peak shear displacement can be predicted using Eq. (1). According to Barton's (1973) model, the tangent dilation angle at peak shear displacement should be as follows:

$$d_{t,peak} = JRC_{peak}\cdot\log\left(\frac{JCS}{\sigma_n}\right) \tag{24}$$

The MDST database contains the results of 341 direct shear tests for which all required information (*JRC*, *JCS*, and normal stress) is available in order to calculate $d_{t,peak}$. The ratio of $d_{t,peak}$ determined using Eq. (23) to $d_{t,peak}$ obtained using Eq. (24) (the peak secant dilation angles were calculated employing Eqs. (10) and (18)) ranged from 0.2 to 1.31 and had an average of 0.82 with standard deviation of 0.2. Therefore, both Eqs. (23) and (24) properly define the same parameter (peak tangent dilation angle).

Substituting Eq. (24) into Eq. (23), the secant dilation angle at peak can be calculated using the following equation:

$$d_{s,peak} = \arctan\left(\frac{1}{3}\tan\left(JRC\cdot\log\left(\frac{JCS}{\sigma_n}\right)\right)\right) \tag{25}$$

As a result, an empirical model is proposed for pre-peak dilatancy behavior of rock fractures. The model is depicted in Fig. 11. The dilation displacement can be calculated at each shear displacement using the following equation:

$$\frac{\delta_v}{\delta_{peak}} = \frac{1}{3}\tan\left(JRC\cdot\log\left(\frac{JCS}{\sigma_n}\right)\right)\cdot\left(\frac{\delta_h}{\delta_{peak}}\right)\cdot\left\{2\left(\frac{\delta_h}{\delta_{peak}}\right)-1\right\} \tag{26}$$

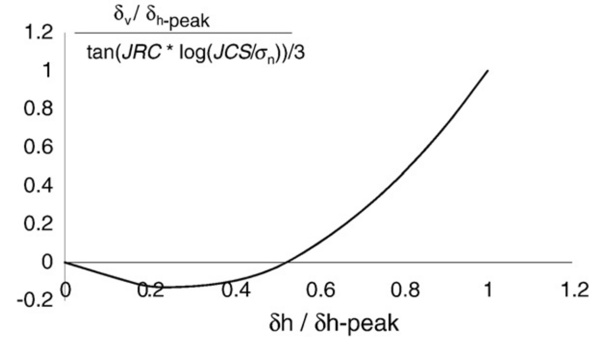

Fig. 11. Proposed model for pre-peak dilation displacement of rock fractures (Eq. (26)).

The proposed model has none of the inconsistencies and ambiguity of Barton's model described in Section 3. Moreover, it simulates negative dilation, while Barton's model does not. For category 2 of the MDST database, Eq. (26) gives $\left(\frac{(\delta_{v,min})_{predicted}}{(\delta_{v,min})_{measured}}\right)_{average} = 0.65$, compared to Barton's model that gave zero. Furthermore, the proposed model predicts zero dilation at $\delta_h/\delta_{peak} = 0.50$, which is somewhat closer to the average measured value for the whole MDST database (Categories 1 and 2), $\delta_h/\delta_{peak} = 0.43$, compared to what Barton's model suggests, $\delta_h/\delta_{peak} = 0.3$. In addition, the dilation displacement at each shear displacement can be calculated easily using Eq. (26); its numerical implementation is also much easier.

3.3.2. Post-peak dilatancy

The tangent dilation angle was defined in Eq. (21). On the other hand, from an analytical point of view, the tangent dilation angle at each shear displacement should be obtained from $JRC_{mobilized}$ using Eq. (24). Substituting Eq. (21) in Eq. (24), we have:

$$d\left(\frac{\delta_v}{\delta_{peak}}\right) = \tan\left(JRC_{mobilized}\log\left(\frac{JCS}{\sigma_n}\right)\right)\cdot d\left(\frac{\delta_h}{\delta_{peak}}\right) \tag{27}$$

Section 3.3 proposed Eq. (17) to predict post-peak mobilized *JRC*. Substituting, Eq. (17) in Eq. (27), we have:

$$d\left(\frac{\delta_v}{\delta_{peak}}\right) = \tan\left(JRC_{peak}\cdot\left(\frac{\delta_{peak}}{\delta_h}\right)^{0.381}\cdot\log\left(\frac{JCS}{\sigma_n}\right)\right)\cdot d\left(\frac{\delta_h}{\delta_{peak}}\right) \tag{28}$$

Dilation displacement between the peak shear displacement and a post-peak point with $\frac{\delta_h}{\delta_{peak}} = l$ can be estimated by obtaining the integral of the right hand side of Eq. (28) between 1 and *l*:

$$\delta_v = \delta_{peak}\cdot\left\{\int_1^l \tan\left(JRC_{peak}\cdot\left(\frac{\delta_{peak}}{\delta_h}\right)^{0.381}\cdot\log\left(\frac{JCS}{\sigma_n}\right)\right)\cdot d\left(\frac{\delta_h}{\delta_{peak}}\right)\right\} + \delta_{v,peak} \tag{29}$$

The integral part of Eq. (29) can be solved by employing numerical methods (for example 1/3-Simpson's rule).

The MDST database contains the post-peak dilation displacements of 205 direct shear tests, for which post-peak dilation at 1 to 4 points are available (total number of 700 data points). Table 6 summarizes the comparison between predicted and measured values. It is found that Eq. (28) works very well for this database.

3.4. Pre-peak stress–displacement curve

As mentioned in Section 2.3, if one wants to use Eq. (1) to describe the pre-peak shear stress–displacement curve, then *JRC* and ϕ_r must depend on the displacement δ. This dependency is called "mobilization" of *JRC* and ϕ_r, respectively.

Table 6
Performance of proposed model, Eq. (29), in predicting the post-peak dilation displacements.

Parameter	Eq. (29)
$\left(\frac{(\delta_v)_{predicted}}{(\delta_v)_{measured}}\right)_{Average} \pm \text{STD}$	1.44 ± 1.31
$\left(\frac{(\delta_v)_{predicted}}{(\delta_v)_{measured}}\right)_{max}$	10.49
$\left(\frac{(\delta_v)_{predicted}}{(\delta_v)_{measured}}\right)_{min}$	0.13

Barton expressed the pre-peak stress–displacement curve by using the concept of roughness mobilization, $JRC_{mobilized}$, in Eq. (1) (Barton, 1982). The ratio $\frac{JRC_{mobilized}}{JRC_{peak}}$ can be estimated from the ratio δ_h/δ_{peak} employing the values given in Table 1. In Barton's model, it was assumed, first, that the base friction angle of rock fracture is mobilized and reaches its peak value at shear displacement of $0.3\delta_{peak}$. Then, from $0.3\delta_{peak}$ to the peak shear displacement, *JRC* is mobilized and reaches its peak value at δ_{peak}. This assumption is consistent with the zero peak shear displacement for a sawed fracture. However, Section 3.2 showed that the peak shear displacement of sawed fractures ($JRC=0$) is significantly different from zero.

The peak shear displacement of lab-size rock blocks is very small, about 1 mm for a 10 cm block. Therefore, for pre-peak shear strength, collecting data by digitizing shear stress versus shear displacement curves published in the literature is very difficult and may lead to large errors. Therefore, in this section, a modification was made to Barton's model for pre-peak shear strength in order to make all parts of the modified model consistent with each other. Accordingly, the mobilization of pre-peak shear strength was divided into two parts: mobilization of friction angle and mobilization of *JRC*.

At zero shear displacement, the shear stress is zero, and thus the mobilized friction angle is zero. At peak shear displacement, the mobilized friction angle is equal to the base friction angle.

For the case of rough fractures ($JRC>0$), at zero shear displacement, the shear stress is zero. Therefore, the mobilized friction angle and *JRC* are both zero. At peak shear displacement, the mobilized friction angle is equal to the base friction angle and the mobilized *JRC* is equal to JRC_{peak}.

Section 3.3.1 suggests empirical Eq. (26) for dilation displacement. Since dilatancy decreases when $0\leq\delta_h\leq0.25\delta_{peak}$ and increases when $\delta_h>0.25\delta_{peak}$, the mobilized *JRC* is assumed to be zero up to $0.25\delta_{peak}$. The mobilized base friction angles at zero and peak shear displacement are known. It was assumed that the shape of shear stress versus shear displacement curve is the same as the linear interpolation used for the results of Barton's model. The MDST database was analyzed in accordance with the above assumptions. Table 7 summarizes the average value of the mobilized base friction angle and the mobilized *JRC*.

4. Validity domains of the modified model

MDST database contained 362 data points of direct shear tests in which the magnitudes of *JRC* ranged between 0 and 20 with a normal distribution. Therefore, the modified model (Eqs. (15), (17), (26), and (29)) is valid for all values of *JRC*.

Table 7
Pre-peak mobilization of the friction angle and *JRC* for stress–displacement calculations.

$\frac{\delta_h}{\delta_{peak}}$	$\frac{\phi_{mobilized}}{\phi_r}$	$\frac{JRC_{mobilized}}{JRC_{peak}}$
0.0	0.0	0.0
0.25	0.75	0
0.5	0.9	0.67
0.6	0.92	0.83
1.0	1.0	1.0

In addition, in MDST database, the σ_n/JCS ratio ranges between 0.001 and 0.1. Consequently, although this range covers almost all stresses that one may be faced in a practical rock engineering problem, it can be considered as a validity domain of the modified model.

Finally, MDST database contained no case with $\delta>10\delta_{peak}$. Although it was demonstrated that Eq. (17) works at least as well as Barton's table (Table 1) for $\delta>10\delta_{peak}$, it is recommended to use Eqs. (15) and (29) with caution at large shear displacements.

5. Conclusions

A database was built by collecting the results of direct shear tests available in the literature: Monotonic Direct Shear Tests (MDST), which contains the results of 366 tests.

Analyses of this database showed that Barton's failure criterion works very well in predicting the shear strength of rock fractures. However, some weaknesses were found in the original Barton model and addressed by correlation analyses performed on collected data. The following modifications to Barton's model are proposed based on the results of correlation analyses:

1) An empirical equation is proposed to predict the peak shear displacement of rock fractures. The equation considers the effect of normal stress on the peak shear displacement, while Barton's equation does not. In addition, this equation can be used for all types of rock fractures, including sawed, smooth, and rough, while Barton's equation predicts an incorrect value of zero for the peak shear displacement of sawed fractures. Finally, the predicted peak shear displacement employing the proposed equation of this study decreases as *JRC* increases. This trend was confirmed by several direct shear tests used to validate the proposed equations (Asadollahi, 2009; Addotto, 2009, Invernizzi, 2009). However, the predicted value of peak shear displacement using Barton's equation increases with *JRC*.
2) An empirical equation is proposed to predict the mobilized *JRC*, which is used to calculate the shear stress–displacement curve after peak shear displacement. Besides better matching the MDST database than Barton's table (Table 1), the empirical equation gives a smoother curve compared to the linear interpolation of the values given in Barton's table and is easier to implement numerically.
3) An equation is proposed to obtain pre-peak dilation at each shear displacement. The proposed model has none of the inconsistencies and ambiguity of Barton's model. Moreover, it simulates negative dilation, while Barton's does not. In addition, the dilation displacement at each shear displacement can be calculated easily using this equation; also the numerical implementation is much easier.
4) An equation is proposed to obtain post-peak dilation at each shear displacement. This equation contains an integral part which should be solved using numerical methods.
5) A table is introduced to simulate the pre-peak shear stress–displacement curve (mobilization of pre-peak shear strength). The table gives the mobilized *JRC* and base friction angle at each shear displacement.

Nomenclature

$d_{s,peak}$	peak secant dilation angle (also called initial dilation angle)
$d_{t,peak}$	peak tangent dilation angle
$d_{t,mobilized}$	mobilized tangent dilation angle
JCS	Joint Compressive Strength
JRC	Joint Roughness Coefficient
L	length of the rock sample in shearing direction
M	damage coefficient
r	Schmidt hammer rebound number on dry unweathered sawed surface
R	Schmidt hammer rebound number on wet joint
δ_h	shear displacement (horizontal displacement)
$\delta_{h,\delta_v=0}$	shear displacement at which dilation is zero

δ_{peak}	peak shear displacement
δ_v	dilation (vertical displacement)
$\delta_{v,min}$	maximum negative dilation
$\delta_{v,peak}$	dilation at peak shear displacement
ϕ_b	base friction angle
ϕ_r	the residual friction angle
π_i	dimensionless parameters
σ_n	normal stress
τ	shear strength of rock fractures
τ_{peak}	peak shear strength of rock fractures

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Appendix A

Monotonic Direct Shear Test Database (MDST):

Studies on shear behavior of rock fractures (Aydan et al., 1990; Bandis, 1980; Bandis et al., 1981; Barton, 1982; Barton et al., 1985; Desai and Fishman, 1991; Grasselli and Egger, 2003; Homand et al., 2001; Huang et al., 2002; Jacobsson, 2004; Jacobsson, 2005a; Jacobsson, 2005b; Jacobsson, 2006a; Jacobsson, 2006b; Jacobsson and Flansbjer, 2005a; Jacobsson and Flansbjer, 2005b; Jacobsson and Flansbjer, 2006a; Jacobsson and Flansbjer, 2006b; Olsson and Barton, 2001; Van Sint Jan, 1990; Wibowo, 1994; Wibowo et al., 1993; Yoshinaka and Yamabe, 1986) were reviewed to find available monotonic direct shear test results, in which the shearing direction is constant (there is no shear unloading and reversal). Peak shear strength (τ_{peak}), peak shear displacement (δ_{peak}), peak positive dilation displacement ($\delta_{v,peak}$), maximum negative dilation ($\delta_{v,min}$), and shear displacement at which dilation displacement is zero ($\delta_{h,\delta_v=0}$) were digitized from the curves. For post-peak behavior, shear strength and dilation displacement at different points were digitized. Tables 8 and 9 presents the MDST database.

A large amount of data was collected from a site investigation report series published by Svensk Karnbranslehantering AB (Swedish Nuclear Fuel and Waste Management Co.) and available online (www.skb.com) (Jacobsson, 2004; Jacobsson, 2005a; Jacobsson, 2005b; Jacobsson, 2006a; Jacobsson, 2006b; Jacobsson and Flansbjer, 2005a; Jacobsson and Flansbjer, 2005b; Jacobsson and Flansbjer, 2006a; Jacobsson and Flansbjer, 2006b). In these cases, the values of *JCS* were assumed to be equal to the unconfined compressive strength of intact rock. In the reports, the peak and residual shear strengths were available in tabular format and the corresponding shear displacements were digitized from curves. Base friction angle and *JRC* values were back calculated assuming that: (1) Barton's (1973) failure criterion can predict the peak and residual shear strength correctly; (2) residual shear strength is reached when $JRC_{mobilized}/JRC_{peak}=0.5$ (Barton, 1982); and (3) no weathering for fractures ($\phi_r=\phi_b$). In order to take into account the effects of multi-stage shearing tests, it was considered that the *JRC* values would be different for the same specimens under different normal stresses (due to the damage of asperities in the shear test run under smaller of normal stresses). The value of the peak dilation displacement, the maximum negative value of dilation, and the shear displacement at which dilation displacement is zero were all digitized from the curves.

Appendix B

The peak shear displacement should decrease by increasing *JRC*:

Fig. 12 depicts a diagram of forces applied in shearing rough fractures. Normal stresses on the horizontal plane, σ, and normal stresses on the inclined plane of fracture, σ_i, can be determined, respectively, as:

$$\sigma = \frac{N}{L} \tag{30}$$

$$\sigma_i = \frac{N_i}{L_i} \tag{31}$$

in which L_i is the length of the fracture along X_i direction and can be obtained from:

$$L_i = L/\cos(i) \tag{32}$$

Therefore, σ_i can be expressed in terms of σ as follows:

$$\sigma_i = \sigma \cdot \frac{\cos(\phi)\cdot\cos(i)}{\cos(\phi+i)} \tag{33}$$

It can be seen in Fig. 12 that the peak shear displacement in X direction, δ, can be expressed in terms of the peak shear displacement in X_i direction, δ_i, as follows:

$$\delta = \delta_i \cdot \cos(i) \tag{34}$$

In order to find whether the peak shear displacement increases or decreases with *JRC*, two fractures are defined with the following conditions (both have the same shape as what is shown in Fig. 12):

– Different dilation angle: $0<i_1<i_2<90$; and thus:

$$JRC_1\cdot\log\left(\frac{JCS}{\sigma_1}\right)<JRC_2\cdot\log\left(\frac{JCS}{\sigma_2}\right) \tag{35}$$

– Equal length along X_i direction: $(L_i)_1=(L_i)_2$
– Equal normal stress on the inclined plane of fracture: $(\sigma_i)_1=(\sigma_i)_2$. Based on Eq. (33) and after simplification, we have:

$$\frac{\sigma_1}{\sigma_2} = \frac{1-\tan(\phi)\cdot\tan(i_1)}{1-\tan(\phi)\cdot\tan(i_2)} \tag{36}$$

While $0<i_1<i_2<90$ and since base friction angles, ϕ, are positive and less than 90°:

$$\frac{\sigma_1}{\sigma_2} = \frac{1-\tan(\phi)\cdot\tan(i_1)}{1-\tan(\phi)\cdot\tan(i_2)}>1 \tag{37}$$

– Both fractures are from the same material and have the same Fracture Compressive Strength (*JCS*) and the same base friction angle (ϕ_b). While *JCS* is a positive value and $JCS>\phi_2,\phi_3$:

$$\log\left(\frac{JCS}{\sigma_1}\right)<\log\left(\frac{JCS}{\sigma_2}\right) \tag{38}$$

Based on Eqs. (35) and (38), it can be concluded:

$$JRC_1<JRC_2 \tag{39}$$

Based on Eq. (34), peak shear displacements have the following relationships:

$$\frac{\delta_1}{\delta_2} = \frac{(\delta_i)_1}{(\delta_i)_2}\cdot\frac{\cos(i_1)}{\cos(i_2)} \tag{40}$$

Table 8
Monotonic Direct Shear Test (MDST) database; Part 1: With post-peak behavior information.

No	L (m)	JRC	JCS (MPa)	δ_{peak} (mm)	σ_n (MPa)	τ_{peak} (MPa)	$\delta_{v,peak}$ (mm)	$\delta_{h,\delta_v=0}$ (mm)	$\delta_{v,min}$ (mm)	Post-peak shear displacements (mm)				Post-peak shear stress (MPa)				Post-peak dilation displacement (mm)			
										1	2	3	4	1	2	3	4	1	2	3	4
Barton (1982); Fig. 5.1; Tension fracture; $\phi_b = 30°$																					
1	0.1	16	0.45	0.33	0.002	–	0.083	0.083	0	1.0	1.5	2.0	3.0	–	–	–	–	0.29	0.38	0.45	0.53
2	0.1	16	0.45	0.46	0.0038	–	0.058	0.300	0	1.0	1.5	2.0	3.0	–	–	–	–	0.21	0.32	0.38	0.49
3	0.1	16	0.45	0.50	0.008	–	0.067	0.300	0	1.0	1.5	2.0	3.0	–	–	–	–	0.17	0.27	0.35	0.44
4	0.1	16	0.45	0.53	0.013	–	0.062	0.300	0	1.0	1.5	2.0	3.0	–	–	–	–	0.13	0.21	0.27	0.35
5	0.1	16	0.45	0.58	0.020	–	0.026	0.300	0	1.0	1.5	2.0	3.0	–	–	–	–	0.15	0.23	0.28	0.33
6	0.1	16	0.45	1.18	0.042	–	0.088	0.710	0	1.5	2.0	2.3	2.5	–	–	–	–	0.17	0.19	0.20	0.21
Bandis (1980); $\phi_b = 32°$																					
7	0.06	15.0	2.00	0.72	0.0245	0.044	–	–	–	1.0	2.0	3.0	5.0	0.035	0.032	0.031	0.029	–	–	–	–
8	0.12	12.2	1.46	0.92	0.0245	0.035	–	–	–	1.0	2.0	3.0	6.0	0.031	0.028	0.027	0.027	–	–	–	–
9	0.18	10.8	1.22	0.97	0.0245	0.029	–	–	–	1.0	2.0	3.0	6.0	0.026	0.026	0.026	0.024	–	–	–	–
10	0.36	8.8	0.89	1.85	0.0245	0.023	–	–	–	1.0	2.0	3.0	6.0	0.024	0.023	0.023	0.023	–	–	–	–
Barton (1982); Fig. 6.2; $\phi_b = 32°$																					
11	0.06	12.5	9.55	0.78	0.0245	0.050	0.22	0.09	0	2.0	3.0	4.0	5.0	0.047	0.044	0.039	0.037	0.41	0.60	0.78	0.97
12	0.12	11.5	4.03	1.25	0.0245	0.040	0.17	0.36	0	2.0	3.0	4.0	5.0	0.038	0.035	0.032	0.031	0.27	0.44	0.66	0.86
13	0.18	11.0	2.31	1.80	0.0245	0.035	0.18	0.73	0	2.0	3.0	5.0	6.0	0.034	0.034	0.032	0.032	0.27	0.44	0.72	0.83
14	0.36	9.0	1.68	2.22	0.0245	0.032	0.11	0.93	0	3.0	4.0	5.0	6.0	0.031	0.040	0.040	0.039	0.25	0.42	0.57	0.65
Barton et al. (1985); Fig. 20; $\phi_b = 30°$																					
15	0.25	8.3	57	1.64	6	4.17	0.07	0.74	0	3.0	6.0	12.0	21.0	3.98	3.79	3.63	3.5	0.15	0.31	0.56	0.85
16	0.25	8.3	57	1.61	24	13. 41	0.03	0.8	0	3.0	6.0	12.0	21.0	13.13	12.97	12.73	12.56	0.05	0.11	0.22	0.33
17	0.75	6.7	41	2.76	6	3.72	0.08	1.01	0	3.0	6.0	12.0	21.0	3.67	3.6	3.52	3.38	0.08	0.21	0.43	0.68
18	0.75	6.7	41	3.07	24	12.57	0.03	1.08	0	3.0	6.0	12.0	21.0	12.54	12.52	12.43	12.26	0.02	0.06	0.06	0.19
Barton et al. (1985); Fig. 22; $\phi_b = 30°$																					
19	0.30	8.0	72	1.55	1	1.02	0.076	0.83	0	4.0	8.0	14.0	22.0	0.91	0.76	0.88	0.86	0.38	0.76	1.23	1.71
20	0.30	8.0	72	1.71	3	2.57	0.061	0.83	0	4.0	8.0	14.0	22.0	0.231	2.22	2.24	2.24	0.28	0.56	0.91	1.27
21	0.30	8.0	72	1.67	10	7.53	0.039	0.83	0	4.0	8.0	14.0	22.0	0.71	6.94	6.86	6.77	0.17	0.34	0.56	0.79
22	0.30	8.0	72	1.67	30	19.67	0.012	0.83	0	4.0	8.0	14.0	22.0	19.31	19	18.86	18.81	0.07	0.15	0.26	0.35
Barton et al. (1985); Fig. 23; $\phi_b = 30°$																					
23	0.10	10	100	0.84	10	8.44	0.04	0.42	0	8.0	14.0	18.0	24.0	7.66	7.49	7.39	7.34	0.45	0.66	0.81	0.97
24	0.30	8.0	72	1.81	10	7.62	0.06	0.77	0	8.0	14.0	18.0	24.0	7.49	7.31	7.20	7.16	0.34	0.55	0.67	0.81
25	1.00	6.3	50	3.61	10	7.05	0.08	1.35	0	8.0	14.0	18.0	24.0	7.21	7.13	7.08	7	0.22	0.39	0.49	0.62
26	3.00	5.1	36	7.34	10	6.64	0.02	2.55	0	8.0	14.0	18.0	24.0	7	6.93	6.86	6.86	0.12	0.25	0.33	0.43
Olsson and Barton (2001); Fig. 16; $\phi_b = 31°$																					
27	0.20	9.7	169	0.87	2	2.35	−0.02	0.93	−0.03	1.0	2.5	4.0	5.5	2.23	1.89	1.77	1.74	0.02	0.4	0.7	0.95
Jacobsson (2005a) and Jacobsson and Flansbjer (2005b); KLX06A; Avro granite; $\phi_b = 33°$																					
28	0.06	12.0	184	0.13	0.5	1.02	0.04	0.04	0.00	0.6	1.0	1.7	2.0	0.8	0.69	0.61	0.57	0.23	0.36	0.52	0.58
29	0.06	7.4	184	0.6	5	4.93	0.065	0.031	−0.02	1.0	2.0	3.2	3.5	4.87	4.45	3.97	3.9	0.14	0.3	0.43	0.46
30	0.06	6.5	184	2.17	20	16.37	0.06	1.18	−0.05	2.6	3.2	4.6	5.0	16.04	15.18	14.19	14.41	0.09	0.11	0.14	0.14
31	0.06	7.7	184	0.24	0.5	0.66	0.03	0.12	0.00	0.5	1.0	1.6	2.0	0.55	0.45	0.45	0.45	0.11	0.21	0.31	0.35
32	0.06	6.6	184	0.3	5	4.71	0.015	0.26	−0.02	1.0	2.0	2.7	3.0	4.32	4.17	4.15	4.13	0.12	0.23	0.29	0.31
33	0.06	6.1	184	2.26	20	16.14	0.09	0.79	−0.04	3.0	4.0	4.5	5.0	1.52	15.06	14.58	14.23	0.12	0.15	0.16	0.16
34	0.06	12.5	184	0.47	0.5	1.07	0.14	0	0.00	1.0	1.4	1.7	2.0	0.77	0.73	0.67	0.62	0.32	0.41	0.5	0.65
35	0.06	7.3	184	0.47	5	4.91	0	0.46	−0.02	1.0	2.0	2.8	3.0	4.73	4.27	4	4	0.1	0.26	0.34	0.37
36	0.06	3.8	184	1.48	20	14.91	−0.01	1.66	−0.07	2.0	3.0	4.5	5.0	15.06	14.46	13.84	13.58	0.02	0.06	0.06	0.05
37	0.06	10.8	184	0.11	0.5	0.89	0.05	0.03	0.00	0.6	1.0	1.7	2.0	0.66	0.57	0.52	0.5	0.15	0.24	0.38	0.42

(continued on next page)

Table 8 (*continued*)

No	*L* (m)	*JRC*	*JCS* (MPa)	δ_{peak} (mm)	σ_n (MPa)	τ_{peak} (MPa)	$\delta_{v,peak}$ (mm)	$\delta_{h,\delta_v=0}$ (mm)	$\delta_{v,min}$ (mm)	Post-peak shear displacements (mm)				Post-peak shear stress (MPa)				Post-peak dilation displacement (mm)			
										1	2	3	4	1	2	3	4	1	2	3	4
Jacobsson (2005a) and Jacobsson and Flansbjer (2005b); KLX06A; Avro granite; $\phi_b=33°$																					
38	0.06	8.7	184	0.26	5	5.3	0.02	0.19	−0.01	1.0	2.0	2.7	3.0	4.6	4	3.9	3.9	0.11	0.24	0.32	0.35
39	0.06	5.7	184	0.54	20	15.9	−0.03	0.9	−0.05	1.0	3.0	4.4	5.0	15.35	14.46	14.12	13.94	0.01	0.1	0.11	0.1
40	0.06	10.7	184	0.3	0.5	0.88	0.055	0.14	−0.01	0.7	1.0	1.7	2.0	0.69	0.63	0.61	0.61	0.1	0.24	0.36	0.42
41	0.06	8.4	184	0.38	5	5.2	0	0.39	−0.03	1.0	2.0	2.7	3.0	4.46	4.56	4.29	4.28	0.1	0.23	0.29	0.31
42	0.06	6.4	184	0.71	20	16.28	−0.01	0.95	−0.03	1.0	3.0	4.4	5.0	15.84	15.36	14.15	14	0	0.1	0.11	0.11
43	0.07	6.3	184	0.88	0.5	0.58	0.13	0	0.00	1.0	1.2	1.7	2.0	0.59	0.53	0.52	0.5	0.14	0.17	0.2	0.23
44	0.07	3.1	184	1.5	5	3.89	0.045	0.46	−0.02	2.0	2.3	3.0	3.2	3.73	3.72	3.53	3.55	0.06	0.08	0.1	0.1
45	0.07	1.4	184	2.87	20	13.67	−0.055	–	−0.06	3.0	4.0	4.4	5.0	13.77	13.28	13.22	13.41	−0.06	−0.06	−0.05	−0.05
Jacobsson (2006b) and Jacobsson and Flansbjer (2005b); KLX10-117; Avro granite; $\phi_b=35°$																					
46	0.06	9.1	172	0.79	0.5	0.8	0.11	0.17	−0.01	1.0	1.3	1.7	2.0	0.69	0.69	0.64	0.64	0.21	0.25	0.3	0.34
47	0.06	6.7	172	0.6	5	5.040	0.005	0.59	−0.03	1.0	2.0	2.7	3.0	4.99	4.67	4.6	4.36	0.58	0.76	0.85	0.89
48	0.06	4.6	172	2.37	20	16.38	0.13	1.16	−0.05	3.0	4.0	4.5	5.0	15.43	14.51	14.14	13.39	0.69	0.73	0.74	0.74
49	0.06	5.3	172	1.46	5	4.68	0.19	0.34	−0.02	2.0	2.7	3.0	–	4.19	4.05	4.08	–	0.26	0.34	0.38	–
50	0.06	2.1	172	4.16	20	15.04	0.27	0.95	−0.05	4.5	5.0	–	–	14.86	14.69	–	–	0.29	0.32	–	–
51	0.06	10.6	172	1.26	0.5	0.94	0.02	0.06	−0.02	1.7	2.0	–	–	0.79	0.78	–	–	0.05	0.07	–	–
52	0.06	3.1	172	3.13	20	15.58	0.11	0.095	−0.04	4.0	4.4	5.0	–	14.89	14.84	14.65	–	0.11	0.1	0.1	–
53	0.07	7.8	172	1.15	0.5	0.71	0.29	0.08	0.00	1.7	2.0	–	–	0.65	0.65	–	–	0.41	0.46	–	–
54	0.07	5.3	172	0.35	5	4.68	0.08	0.23	−0.01	1.0	2.0	2.7	3.0	4.34	4.33	4.11	4.04	0.16	0.37	0.45	0.49
55	0.07	4.3	172	0.63	20	16.18	0	0.61	−0.30	2.0	4.0	4.4	5.0	14.84	14.44	14.39	13.79	0.17	0.34	0.36	0.41
56	0.06	9.9	172	0.45	0.5	0.87	0.06	0.19	0.00	1.0	1.7	2.0	–	0.71	0.64	0.56	–	0.26	0.45	0.51	
57	0.06	7.8	172	0.58	5	5.35	0.01	0.48	−0.02	1.0	2.0	2.7	3.0	4.98	4.46	4.17	4.07	0.11	0.33	0.43	0.47
58	0.06	4.4	172	0.94	20	16.28	−0.03	1.29	−0.05	2.0	4.0	4.5	5.0	15.01	13.96	13.71	13.06	0.04	0.05	0.04	0.02
Jacobsson (2006b) and Jacobsson and Flansbjer (2006b); KLX12A-117; Avro granite; $\phi_b=35°$																					
59	0.07	10.9	175	0.13	0.5	0.97	0.04	0.02	0.00	1.0	1.7	2.0	–	0.8	0.82	0.79	–	0.32	0.49	0.55	-
60	0.07	7.3	175	2.38	5	5.22	0.44	0.09	−0.01	2.7	3.0	–	–	4.93	4.75	–	–	0.5	0.55	–	–
61	0.07	5.6	175	2.68	20	16.95	0.11	0.73	−0.02	3.0	4.0	4.5	5.0	16.44	15.61	15.4	14.7	0.13	0.16	0.16	0.17
Jacobsson (2006b) and Jacobsson and Flansbjer (2006b); KLX12A-117; Avro granite; $\phi_b=34°$																					
62	0.06	11.8	268.7	0.71	0.5	1.14	0.2	0.15	0.00	1.0	1.8	2.0	–	1.04	0.91	0.79	–	0.3	0.48	0.51	–
63	0.06	6.5	268.7	0.76	5	5.03	0.04	0.56	−0.04	1.0	2.0	2.8	3.0	4.92	4.7	4.56	4.54	0.09	0.25	0.36	0.38
64	0.06	3.9	268.7	2.73	20	15.88	0.05	1.13	−0.05	3.0	4.0	4.5	5.0	15.2	13.68	13.63	13.37	0.07	0.05	0.04	0.03
65	0.06	13.4	268.7	0.63	0.5	1.42	0.2	0.1	0.00	1.0	1.7	2.0	–	1.14	0.93	0.85	–	0.32	0.46	0.52	–
66	0.06	6.4	268.7	0.34	5	5	−0.01	0.37	−0.02	1.0	2.0	2.7	3.0	4.33	4.12	4.23	4.28	0.11	0.19	0.22	0.24
67	0.06	2.4	268.7	3.13	20	14.91	−0.02	–	–0.04	4.0	4.6	5.0	–	14.73	14.59	14.47	–	−0.04	−0.05	−0.06	–
Jacobsson (2006a) and Jacobsson and Flansbjer (2006b); KFM09A-117; Medium-grained granite; $\phi_b=35°$																					
68	0.06	8.4	250	0.49	0.5	0.79	0.09	0.13	0.00	1.0	1.7	2.0	–	0.61	0.65	0.63	–	0.24	0.4	0.46	–
69	0.06	6.0	250	1.69	5	5.04	0.2	0.46	−0.04	2.0	2.7	3.0	–	4.54	4.41	4.33	–	0.24	0.31	0.34	–
70	0.06	3.9	250	3.12	20	16.37	−0.02	4.2	−0.09	4.0	4.4	5.0	–	15.37	15.15	14.97	–	0	0	−0.01	–
71	0.05	9.3	250	0.18	0.5	0.87	0.03	0.1	0.00	1.0	1.7	2.0	–	0.61	0.61	0.58	–	0.26	0.39	0.45	–
72	0.05	6.2	250	0.42	5	5.1	−0.03	0.56	−0.04	1.0	2.0	2.7	3.0	4.5	3.88	3.66	3.49	0.08	0.21	0.28	0.31
73	0.05	2.0	250	1.09	20	15.19	−0.1	–	−0.11	2.0	4.0	4.4	5.0	14.11	13.3	12.97	12.83	−0.05	−0.04	−0.04	−0.07
74	0.05	6.6	250	0.47	0.5	0.66	0.07	0.3	−0.01	1.0	1.7	2.0	–	0.58	0.5	0.44	–	0.22	0.39	0.45	–
75	0.05	3.1	250	0.57	5	4.24	−0.01	0.62	−0.04	1.0	2.0	2.7	3.0	3.92	3.95	4.1	4.08	0.07	0.22	0.31	0.35
76	0.05	1.5	250	4.13	20	14.9	−0.06	–	−0.12	4.4	5.0	–	–	13.3	13.27	–	–	−0.07	−0.08		–
77	0.05	9.4	250	0.18	0.5	0.88	0.08	0	0.00	1.0	1.7	2.0	–	0.53	0.52	0.44	–	0.23	0.33	0.35	–
78	0.05	3.6	250	1.25	5	4.37	0.11	0.33	−0.02	2.0	2.7	3.0	–	3.84	3.76	3.71	–	0.18	0.23	0.25	–
79	0.05	1.1	250	3.62	20	14.62	0.02	2.51	−0.07	4.0	4.5	5.0	–	13.76	14	13.77	–	0.02	0	0	–
Jacobsson (2005a) and Jacobsson and Flansbjer (2005a); KFM06A-117; Medium-grained granite; $\phi_b=30°$																					
80	0.06	13.4	185	0.11	0.5	1.04	0.04	0.03	0.00	1	1.69	2	–	0.57	0.49	0.442	–	0.28	0.38	0.42	–
81	0.06	8.8	185	0.23	5	4.79	0.01	0.13	0.00	1	2	2.71	3	3.981	3.48	3.36	3.374	0.13	0.23	0.28	0.3

82	0.06	5.4	185	0.39	20	14.13	0	0.39	−0.09	2	4	4.71	5	13.62	12.93	12.72	12.45	0.11	0.19	0.21	0.22
83	0.06	10.2	185	0.21	0.5	0.75	0.04	0.08	0.00	1	1.67	2	–	0.5	0.53	0.439	–	0.25	0.4	0.45	–
84	0.06	7.8	185	1.52	5	4.53	0.24	0.25	−0.01	2	2.38	–	–	3.91	4.05	–	–	0.31	0.36	–	–
85	0.06	7.5	185	0.46	20	15.2	−0.02	0.64	−0.03	2	4	4.44	5	14.37	13.83	13.9	13.67	0.12	0.19	0.2	0.22
86	0.06	5.2	185	1.14	0.5	0.47	0.12	0.27	0.00	1.7	2	–	–	0.45	0.441	–	–	0.21	0.25	–	–
87	0.06	4.8	185	1.53	5	3.84	0.1	0.49	−0.01	1	2.87	3	–	3.70	3.5	3.564	–	0.14	0.2	0.21	–
88	0.06	5.5	185	5.05	20	14.16	0.06	2.22	−0.04	5.1	–	–	–	14.16	–	–	–	0.06	–	–	–
89	0.06	8.6	185	0.31	0.5	0.64	0.05	0.04	0.00	1	1.72	2	–	0.441	0.42	0.441	–	0.15	0.26	0.3	–
90	0.06	5.9	185	0.44	5	4.09	−0.01	0.5	−0.02	1	2	3.01	–	3.92	3.904	3.82	–	0.03	0.08	0.12	–
91	0.06	5.9	185	2.56	20	14.35	−0.13	–	−0.23	3.0	4.0	4.5	5.0	13.99	13.64	13.65	13.50	−0.14	−0.18	−0.2	−0.23
92	0.06	11.4	185	0.1	0.5	0.84	0.005	0.04	0.00	1.0	1.7	2.0	–	0.592	0.59	0.554	–	0.24	0.38	0.46	–
93	0.06	8.5	185	0.29	5	4.72	−0.01	0.35	−0.02	1.0	2.0	2.7	3.0	4.319	4.276	4.38	4.281	0.09	0.22	0.28	0.3
94	0.06	9.5	185	2.24	20	16.29	0	2.19	−0.08	3.0	4.0	4.5	5.0	15.49	15.11	15.22	14.92	0.03	0.06	0.06	0.06
Jacobsson (2005a) and Jacobsson and Flansbjer (2005a); KFM06A-117; Medium-grained granite; $\phi_b = 30°$																					
95	0.07	13.4	185	0.15	0.5	1.04	0.01	0.12	0.00	1.0	1.7	2.0	–	0.554	0.5	0.491	–	0.27	0.39	0.44	–
96	0.07	12.1	185	0.32	5	5.75	−0.02	0.39	−0.03	1.0	2.0	2.9	3.0	4.391	4.278	3.8	3.574	0.11	0.25	0.33	0.34
97	0.07	10.6	185	0.98	20	16.9	−0.06	1.71	−0.08	2.0	4.0	4.7	5.0	15.28	13.59	13.02	12.73	0.02	0.12	0.13	0.13
Jacobsson (2005a) and Jacobsson and Flansbjer (2005a); KFM05A-117; Medium-grained granite; $\phi_b = 29°$																					
98	0.05	15.5	230	0.12	0.5	1.33	0.02	0.06	0.00	1	1.7	2	–	0.608	0.57	0.529	–	0.32	0.46	0.51	–
99	0.05	11.2	230	0.38	5	5.29	0	0.38	−0.02	1	2	2.71	3	4.13	3.81	3.79	3.492	0.13	0.29	0.36	0.39
100	0.05	9.5	230	1.12	20	15.69	−0.08	2.51	−0.09	2	4	4.46	5	14.29	14.34	14.37	13.81	−0.02	0.02	0.02	0.01
101	0.05	14.7	230	0.11	0.5	1.19	0.03	0.02	0.00	1	1.68	2	–	0.608	0.51	0.423	–	0.33	0.5	0.57	–
102	0.05	11.5	230	0.33	5	5.38	0	0.29	−0.01	1	2	2.7	3	4.34	4.021	4.07	4.18	0.15	0.34	0.41	0.45
103	0.05	8.7	230	0.94	20	15.17	−0.07	1.83	−0.08	4	5	4.45	5	14.29	13.81	14.33	14.5	0.01	0.07	0.08	0.09
104	0.06	7.8	230	0.46	0.5	0.57	0.1	0	0.00	1	1.83	2	–	0.423	0.4	0.423	–	0.19	0.29	0.31	–
105	0.06	2.9	230	0.38	5	3.22	−0.005	0.49	−0.02	1	2	2.69	3	2.99	2.778	2.75	2.752	0.03	0.07	0.08	0.09
106	0.06	0.9	230	6.69	20	11.06	−0.05	–	−0.06	7	–	–	–	11.03	–	–	–	−0.06	–	–	–
107	0.06	12.2	230	0.15	0.5	0.88	0.04	0.06	0.00	1	1.67	2	–	0.555	0.5	0.476	–	0.34	0.51	0.59	–
108	0.06	10.6	230	0.24	5	5.1	0	0.25	−0.02	1	2	2.7	3	4.18	3.81	3.47	3.333	0.16	0.3	0.38	0.41
109	0.06	8.1	230	0.47	20	14.84	0	0.52	−0.02	2	4	4.43	5	13.86	12.92	13.16	13.20	0.13	0.25	0.27	0.28
110	0.07	14.2	230	0.15	0.5	1.23	0	0.12	0.00	1	2	2.67	3	0.53	0.68	0.67	0.61	0.32	0.59	0.79	0.89
111	0.07	12.8	230	0.25	5	6.22	0	0.24	−0.01	1	2	2.66	3	4.59	5.06	4.51	4.03	0.21	0.44	0.57	0.63
112	0.07	11.8	230	0.46	20	18.37	0.21	0.77	−0.04	3	4	4.43	5	16.55	16.13	16.6	15.73	0.26	0.31	0.33	0.35
113	0.06	9.5	230	0.21	0.5	0.72	0.03	0.12	0.00	1	1.68	2	–	0.6	0.52	0.48	–	0.2	0.29	0.33	–
114	0.06	7.4	230	0.46	5	4.54	0	0.46	−0.02	1	2	2.86	3	4.18	3.86	3.52	3.46	0.06	0.15	0.23	0.25
115	0.06	5.2	230	3.02	20	14.29	−0.06	0	−0.08	4	5.52	5	–	13.22	13.4	13.06	–	−0.06	−0.07	−0.09	–
116	0.06	13.8	230	0.1	0.5	1.16	0.03	0.05	0.00	1	1.66	2	–	0.596	0.55	0.546	–	0.34	0.48	0.55	–
117	0.06	9.0	230	0.28	5	4.98	0.02	0.14	0.00	1	2	2.64	3	4.159	3.816	3.66	3.6	0.19	0.34	0.41	0.44
118	0.06	6.2	230	0.84	20	14.83	0.01	0.75	−0.03	2	4	4.52	5	13.44	12.05	13.69	12.75	0.12	0.17	0.17	0.16
119	0.06	13.0	230	0.05	0.5	1.06	0.02	0.03	0.00	1	1.66	2	–	0.503	0.48	0.398	–	0.33	0.45	0.5	
120	0.06	10.2	230	0.24	5	5.34	0.01	0.15	0.00	1	2	2.67	3	3.941	3.55	3.52	3.525	0.18	0.32	0.4	0.43
121	0.06	7.7	230	0.58	20	15.72	−0.03	0.96	−0.05	2	4	4.39	5	13.18	12.26	12.92	11.72	0.09	0.16	0.17	0.17
Jacobsson (2005a) and Jacobsson and Flansbjer (2005a); KFM08A-117; Medium-grained granite; $\phi_b = 31°$																					
122	0.06	15.8	173	0.16	0.5	1.46	0.07	0.05	0.00	1	2	2.36	–	0.596	0.47	0.48	–	0.36	0.58	0.64	–
123	0.06	12.0	173	0.34	5	5.86	0.02	0.28	−0.01	1	2	2.76	3	4.212	3.755	3.47	3.378	0.17	0.33	0.42	0.44
124	0.06	6.7	173	0.72	20	15.23	−0.03	0.99	−0.04	2	4	4.61	5	13.63	13.14	13	13.02	0.07	0.12	0.11	0.11
125	0.05	11.0	173	0.13	0.5	0.83	0.03	0.04	0.00	1	1.73	2	–	0.545	0.53	0.52	–	0.31	0.48	0.53	–
126	0.05	8.8	173	0.33	5	4.91	0.02	0.27	−0.01	1	2	2.78	3	4.465	4.389	4.11	4.082	0.17	0.33	0.42	0.44
127	0.05	8.9	173	2.99	20	16.39	0.07	0.99	−0.04	4	4.54	5	–	15.18	15.18	14.41	–	0.06	0.04	0.02	–
128	0.05	15.4	173	0.22	0.5	1.38	0.06	0.12	0.00	1	1.7	2	–	0.697	0.68	0.66	–	0.36	0.54	0.62	–
Jacobsson (2005a) and Jacobsson and Flansbjer (2005a); KFM08A-117; Medium-grained granite; $\phi_b = 31°$																					
129	0.05	12.8	173	0.39	5	6.11	0.02	0.33	−0.02	1	2	2.72	3	5.366	4.503	4.38	4.123	0.19	0.39	0.5	0.54
130	0.05	10.3	173	2.78	20	17.18	0.1	1.26	−0.07	3	4	4.52	5	16.55	15.79	14.95	14.98	0.11	0.15	0.16	0.16
131	0.06	9.1	173	0.3	0.5	0.69	0.06	0.12	0.00	1	1.71	2	–	0.457	0.42	0.444	–	0.25	0.37	0.41	–
132	0.06	7.6	173	0.38	5	4.61	0.03	0.31	−0.02	1	2	2.81	3	3.679	3.514	3.34	3.438	0.15	0.25	0.32	0.33

(continued on next page)

Table 8 (*continued*)

No	L (m)	JRC	JCS (MPa)	δ_{peak} (mm)	σ_n (MPa)	τ_{peak} (MPa)	$\delta_{v,peak}$ (mm)	$\delta_{h,\delta_v=0}$ (mm)	$\delta_{v,min}$ (mm)	Post-peak shear displacements (mm)				Post-peak shear stress (MPa)				Post-peak dilation displacement (mm)			
										1	2	3	4	1	2	3	4	1	2	3	4
Jacobsson (2005a) and Jacobsson and Flansbjer (2005a); KFM08A-117; Medium-grained granite; $\phi_b=31°$																					
133	0.06	4.8	173	0.58	20	14.25	−0.02	0.72	−0.04	2	4	4.56	5	13.31	12.32	12.79	12.42	0.09	0.13	0.13	0.13
134	0.06	10.8	173	0.29	0.5	0.81	0.06	0.15	0.00	1	1.78	2	–	0.672	0.71	0.672	–	0.27	0.46	0.54	–
135	0.06	8.1	173	0.42	5	4.74	0.03	0.32	−0.01	1	2	2.77	3	4.567	4.541	3.98	3.932	0.16	0.35	0.48	0.52
136	0.06	7.0	173	2.48	20	15.40	0.14	0.84	−0.04	3	4	4.72	5	14.55	14.07	14.04	14.10	0.16	0.18	0.2	0.2
Jacobsson (2005b) and Jacobsson and Flansbjer (2005b); KLX03A-117; Avro granite; $\phi_b=34°$																					
137	0.05	6.6	158	1.18	0.5	0.63	0.23	0.04	0.00	1.7	2	–	–	0.58	0.57	–	–	0.32	0.36	–	–
138	0.05	7.1	158	0.99	5	4.93	0.1	0.27	−0.01	2	2.81	3		4.61	4.49	4.39	–	0.24	0.33	0.35	–
139	0.05	7.3	158	1.92	20	17.14	0.02	1.5	−0.05	3	4	4.85	5	16.70	16.56	16.57	16.41	0.07	0.09	0.11	0.11
140	0.06	10.0	158	0.42	0.5	0.83	0.08	0.1	0.00	1	2	2.22	–	0.69	0.57	0.55	0.00	0.17	0.28	0.3	–
141	0.06	4.1	158	1.23	5	4.22	0.03	0.69	−0.02	2	2.72	3	–	4.12	4.02	3.94	–	0.05	0.06	0.07	–
142	0.06	3.2	158	1.79	20	15.01	−0.05	–	−0.17	3	5	5.36	6	14.59	15.77	14.80	14.45	−0.08	−0.14	−0.15	−0.17
143	0.06	6.1	158	0.33	0.5	0.58	0.04	0.05	0.00	1	1.72	2	–	0.50	0.49	0.44	–	0.15	0.23	0.26	–
144	0.06	5.5	158	0.43	5	4.53	0	0.41	−0.01	1	2	2.69	3	4.15	3.83	3.81	3.80	0.15	0.16	0.21	0.23
145	0.06	2.7	158	3.33	20	14.77	0.09	1.18	−0.04	4	4.49	5	–	14.58	14.20	13.70	–	0.11	0.12	0.12	–
146	0.06	7.9	158	0.13	0.5	0.68	0.01	0	0.00	1	1.68	2	–	0.57	0.57	0.55	–	0.2	0.32	0.37	–
147	0.06	4.9	158	1.84	5	4.39	0.19	0.33	−0.01	2	3	3.71	4	4.19	3.58	3.46	3.53	0.21	0.32	0.37	0.39
148	0.06	4.2	158	2.23	20	15.49	0.09	0.56	−0.01	3	4	5	5.51	14.70	14.90	14.91	14.99	0.11	0.12	0.12	0.12
149	0.06	11.4	158	0.18	0.5	0.96	0.02	0	0.00	1	1.68	2	–	0.64	0.55	0.44	–	0.28	0.42	0.48	–
150	0.06	8.4	158	0.4	5	5.29	0.01	0.34	−0.01	1	2	2.7	3	4.70	4.37	3.98	3.95	0.12	0.25	0.32	0.34
151	0.06	4.9	158	0.99	20	15.85	−0.01	1.1	−0.03	2	3	4.45	5	15.69	15.09	14.62	14.46	0.05	0.14	0.15	0.17
152	0.05	7.2	158	1.16	0.5	0.64	0.09	0.35	−0.01	1.8	2	–	–	0.55	0.53	–	–	0.21	0.24	–	–
153	0.05	4.0	158	1.3	5	4.2	0.05	0.77	−0.03	2	2.7	3	–	4.01	3.89	3.90	–	0.12	0.19	0.21	–
154	0.05	1.7	158	5.1	20	14.28	0	4.96	−0.06	–	–	–	–	–	–	–	–	–	–	–	–
155	0.06	6.1	158	0.56	0.5	0.58	0.02	0.37	−0.01	1	1.68	2	–	0.50	0.49	0.44	–	0.11	0.23	0.27	–
156	0.06	3.4	158	0.67	5	4.06	−0.02	0.83	−0.05	1	2	2.71	3	3.95	3.87	3.64	3.52	0.03	0.13	0.18	0.2
157	0.06	1.9	158	1.32	20	14.36	−0.05	2.3	−0.10	2	4	4.6	5	13.69	13.56	13.63	13.49	−0.01	0.06	0.06	0.07
Jacobsson (2005b) and Jacobsson and Flansbjer (2005b); KLX03A-117; Quartz monzodiorite; $\phi_b=31°$																					
158	0.06	12.7	176	0.17	0.5	1	0.03	0.11	−0.01	1	1.87	2	–	0.635	0.56	0.521	–		0.22	0.36	0.39
159	0.06	9.3	176	0.32	5	5.06	0	0.32	−0.02	1	2	2.73	3	4.53	4.08	4.01	3.859	0.09	0.2	0.26	0.29
160	0.06	4.9	176	1.07	20	14.30	−0.04	3.01	−0.05	2	4	4.58	5	14.13	12.66	12.72	12.50	−0.03	−0.01	−0.02	−0.03
Jacobsson (2005a) and Jacobsson and Flansbjer (2005a); KFM07A-117; Medium-grained granite; $\phi_b=35°$																					
161	0.06	9.4	206	0.1	0.5	0.85	0.03	0.05	0.00	1	1.7	2	–	0.647	0.56	0.546	–	0.29	0.42	0.47	–
162	0.06	9.7	206	0.24	5	6.09	0.01	0.2	−0.01	1	2	2.7	3	4.494	4.443	4.37	4.329	0.16	0.29	0.37	0.4
163	0.06	5.1	206	0.5	20	16.89	−0.01	0.66	−0.03	2	4	4.45	5	15.73	15.26	15.01	14.93	0.09	0.17	0.17	0.18
164	0.06	9.4	206	0.07	0.5	0.85	0.02	0.04	0.00	1	1.67	2	–	0.647	0.56	0.546	–	0.25	0.36	0.41	–
165	0.06	9.0	206	0.23	5	5.85	0.01	0.19	−0.01	1	2	2.67	3	4.722	4.595	4.25	4.164	0.14	0.25	0.31	0.33
166	0.06	5.1	206	0.75	20	16.87	−0.01	1.06	−0.03	2	4	4.45	5	16.54	15.53	15.63	15.48	0.02	0.1	0.1	0.1
167	0.06	8.8	206	0.17	0.5	0.8	0.02	0.08	0.00	1	1.68	2	–	0.571	0.57	0.533	–	0.18	0.28	0.33	–
168	0.06	4.8	206	1.11	5	4.61	0.11	0.23	−0.01	2	2.69	3	–	4.329	4.26	4.291	–	0.19	0.24	0.26	–
169	0.06	4.6	206	3.93	20	16.61	0.09	1.1	−0.03	4	4.66	5	–	16.52	16.5	16.52	–	0.09	0.1	0.1	–
170	0.05	8.3	206	0.19	0.5	0.76	0.03	0.08	0.00	1	1.66	2	–	0.609	0.54	0.482	–	0.2	0.27	0.3	–
171	0.05	5.2	206	0.32	5	4.74	−0.005	0.36	−0.02	1	2	2.68	3	4.265	3.986	4	3.935	0.08	0.16	0.19	0.21
172	0.05	3.2	206	1.88	20	15.76	0	1.46	−0.03	3	4	4.52	5	15.44	15.03	14.79	14.29	0.01	0.01	0.01	0.01
173	0.06	8.9	206	0.23	0.5	0.81	0.07	0.03	0.00	1	1.7	2	–	0.597	0.58	0.571	–	0.25	0.37	0.42	–
174	0.06	6.8	206	0.27	5	5.18	0.02	0.18	−0.01	1	2	2.72	3	4.519	4.443	4.47	4.38	0.13	0.24	0.32	0.3
175	0.06	5.8	206	1.85	20	17.29	0.06	0.55	−0.02	3	4	4.45	5	16.85	16.49	16.24	16.07	0.09	0.11	0.12	0.12
Jacobsson (2004); KLX02-117; Avro granite; $\phi_b=34°$																					
176	0.06	5.8	20.23	0.65	0.5	0.57	0.11	0.14	0.00	1	1.73	2	–	0.52	0.47	0.444	–	0.17	0.34	0.4	–
177	0.06	2.3	20.23	1.47	5	3.8	0.13	0.8	−0.02	2	2.9	3	–	3.654	3.59	3.604	–	0.23	0.39	0.43	–
178	0.06	0.9	20.23	2.12	20	13.69	0.11	1.27	−0.03	3	4	4.63	5	13.46	12.95	12.7	12.75	0.23	0.37	0.44	0.49

179	0.06	10.3	20.23	0.48	0.5	0.88	0.11	0.05	0.00	1	1.7	2	–	0.647	0.66	0.609	–	0.28	0.48	0.54	–
180	0.06	9.0	20.23	0.52	5	5.54	0.04	0.31	−0.01	1	3	3.69	4	5.278	4.694	4.47	4.466	0.16	0.52	0.61	0.65
181	0.06	8.2	20.23	0.9	20	17.84	0.02	0.68	−0.03	2	4	4.7	5	16.18	14.88	14.36	14.34	0.16	0.34	0.38	0.41
182	0.05	7.4	20.23	0.22	0.5	0.66	0.02	0.05	0.00	1	1.98	–	–	0.295	0.62	–	–	0.12	0.21	–	–
183	0.05	4.1	20.23	1.93	5	4.22	0.1	0.49	−0.01	3	3.2	–	–	4.009	4.02	–	–	0.15	0.17	–	–
184	0.05	3.6	20.23	3.98	20	15.12	0.07	1.05	−0.02	4.5	5	–	–	14.99	14.71	–	–	0.08	0.08	–	–
185	0.05	12.2	20.23	0.13	0.5	1.09	0.02	0.11	0.00	1	1.7	2	–	0.66	0.53	0.444	–	0.26	0.41	0.46	–
186	0.05	7.7	20.23	0.28	5	5.16	0	0.27	−0.01	1	2	2.71	3	4.225	3.844	3.61	3.578	0.14	0.29	0.35	0.37
187	0.05	5.3	20.23	0.7	20	16.09	0	0.71	−0.03	2	4	4.52	5	13.94	11.10	10.9	10.9	0.11	0.19	0.19	0.19
188	0.06	7.1	20.23	1.1	0.5	0.64	0.07	0	0.00	1.9	2.22	–	–	0.59	0.596	–	–	0.14	0.17	–	–
189	0.06	6.4	20.23	3.11	5	4.79	0.21	0.93	−0.02	3.7	4	–	–	4.48	4.377	–	–	0.28	0.33	–	–
190	0.06	7.8	20.23	2.95	20	17.57	0.1	1.76	−0.05	4	4.95	5.55	–	15.78	15.48	15.72	–	0.19	0.24	0.28	–
191	0.05	12.7	20.23	0.1	0.5	1.15	0.03	0	0.00	1	1.92	2	–	0.634	0.58	0.533	–	0.29	0.44	0.46	–
192	0.05	7.4	20.23	0.26	5	5.06	0.01	0.21	−0.02	1	2	3.18	–	4.352	3.743	3.05	0	0.13	0.24	0.33	0.34
193	0.05	4.8	20.23	0.48	20	15.79	0	0.49	−0.01	2	4	4.64	5	12.97	12.6	12.18	11.94	0.12	0.2	0.2	0.2
194	0.06	10.5	20.23	0.47	0.5	0.9	0.11	0.1	0.00	1	1.68	2	–	0.749	0.63	0.596	–	0.28	0.46	0.53	–
195	0.06	9.8	20.23	0.49	5	5.79	0.02	0.41	−0.02	1	2	2.75	3	5.189	4.223	3.86	3.73	0.13	0.33	0.47	0.51
196	0.06	7.7	20.23	1.09	20	17.51	0.02	0.87	−0.04	2	4	4.44	5	16.63	13.97	14.24	14.91	0.12	0.3	0.32	0.34
Jacobsson (2004); KLX02-117; Avro granite; $\phi_b=34°$																					
197	0.06	11.4	20.23	0.14	0.5	0.99	0.03	0.08	0.00	1	2	2.48	2.81	0.787	0.647	0.58	0.609	0.29	0.49	0.57	0.63
198	0.06	8.7	20.23	0.4	5	5.46	0.01	0.32	−0.02	1	3	3.65	4	5.188	4.365	4.18	4.06	0.1	0.38	0.46	0.49
199	0.06	7.3	20.23	2.66	20	17.29	0.09	1.09	−0.04	3	4	4.96	5.53	17.10	16.66	16.15	16.05	0.1	0.12	0.12	0.12
200	0.06	4.6	20.23	1.09	0.5	0.51	0.07	0.13	0.00	1.7	2	–	–	0.46	0.444	–	–	0.14	0.16	–	–
201	0.06	2.0	20.23	4.54	5	3.73	0.26	1.19	−0.03	–	–	–	–	–	–	–	–	–	–	–	–
202	0.06	0.8	20.23	5.68	20	13.66	0.12	2.94	−0.04	6	–	–	–	13.51	–	–	–	0.13	–	–	–
Homand et al. (2001); Schist replicas; $\phi_b=34°$																					
203	0.15	11.0	75	8.16	0.4	0.64	1.17	3	0.00	10	15	20	–	0.562	0.454	0.432	–	1.7	2.57	3.14	–
204	0.15	11.0	75	9.63	0.8	1.15	2.12	1	0.00	10	15	20	–	1.119	0.88	0.78	–	2.27	3.4	4.13	–
205	0.15	11.0	75	8.17	1.2	1.67	1.56	0.25	0.00	10	15	20	–	1.492	1.416	1.265	–	1.94	2.96	3.78	–
206	0.15	11.0	75	7.68	1.8	2.36	1.35	0.5	0.00	10	15	20	–	2.189	2.005	1.789	–	2.04	2.96	3.46	–
207	0.15	11.0	75	12.65	2.4	2.87	1.17	5.31	−0.18	15	20	–	–	2.47	2.486	–	–	1.49	2.03	–	–
Bandis et al. (1981); Fig. 4; Tension fracture; $\phi_b=32°$																					
208	0.09	6.5	2	0.47	0.01	0.0113	–	–	–	2	3	4	6	0.012	0.012	0.012	0.011	–	–	–	–
209	0.09	7.5	2	0.55	0.01	0.015	–	–	–	2	3	4	6	0.015	0.015	0.014	0.012	–	–	–	–
210	0.09	10.6	2	0.73	0.01	0.017	–	–	–	2	3	4	6	0.017	0.016	0.015	0.014	–	–	–	–
211	0.09	16.6	2	0.73	0.01	0.031	–	–	–	2	3	4	6	0.024	0.022	0.019	0.017	–	–	–	–
212	0.09	6.5	2	0.56	0.034	0.032	–	–	–	2	3	5	7	0.033	0.031	0.030	0.028	–	–	–	–
213	0.09	7.5	2	0.61	0.034	0.036	–	–	–	2	3	5	7	0.035	0.034	0.031	0.028	–	–	–	–
214	0.09	10.6	2	0.8	0.034	0.041	–	–	–	2	3	5	7	0.040	0.038	0.035	0.033	–	–	–	–
215	0.09	16.6	2	0.85	0.034	0.053	–	–	–	2	3	5	7	0.045	0.043	0.040	0.037	–	–	–	–
216	0.09	6.5	2	1.18	0.09	0.072	–	–	–	2	4	6	7	0.071	0.069	0.068	0.067	–	–	–	–
217	0.09	7.5	2	1.07	0.09	0.082	–	–	–	2	4	6	7	0.079	0.075	0.072	0.071	–	–	–	–
218	0.09	10.6	2	0.99	0.09	0.091	–	–	–	2	4	6	7	0.088	0.083	0.078	0.077	–	–	–	–
219	0.09	16.6	2	1.33	0.09	0.123	–	–	–	2	4	6	7	0.119	0.110	0.105	0.102	–	–	–	–
Bandis et al. (1981); Figs. 9–21; Bedding plane in limestone; $\phi_b=32°$																					
220	0.06	17.5	2000	0.57	0.0245	0.057	0.15	0.2	0.00	2	3	4	–	0.048	0.045	0.042	–	0.39	0.55	0.66	–
221	0.12	14.2	1375.5	0.74	0.0245	0.052	0.14	0.2	0.00	2	3	4	–	0.046	0.042	0.040	–	0.41	0.63	0.79	–
222	0.18	12.6	1105.1	1.17	0.0245	0.042	0.1	0.4	0.00	2	3	4	6	0.039	0.039	0.036	0.034	0.28	0.48	0.63	0.96
223	0.36	10.2	760.03	2.7	0.0245	0.036	0.16	0.8	0.00	3	4	6	–	0.036	0.036	0.035	–	0.24	0.45	0.8	–
Bandis et al. (1981); Figs. 9–21; Bedding plane in slightly metamorphosed fine grained sandstone; $\phi_b=32°$																					
224	0.06	17.5	2	0.75	0.0245	0.022	0.02	0.39	0.00	2	3	4	5	0.022	0.021	0.021	0.021	0.06	0.09	0.11	0.13
225	0.12	14.2	1.799	1.15	0.0245	0.021	0.03	0.39	0.00	2	3	4	5	0.020	0.020	0.020	0.020	0.06	0.08	0.11	0.12
226	0.18	12.6	1.691	1.1	0.0245	0.020	0.02	0.39	0.00	2	3	4	5	0.020	0.019	0.019	0.019	0.04	0.07	0.1	0.11
227	0.36	10.2	1.521	1.25	0.0245	0.019	0	0.94	0.00	2	3	4	5	0.019	0.019	0.018	0.019	0.03	0.05	0.08	0.09

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Table 8 (*continued*)

No	*L* (m)	*JRC*	*JCS* (MPa)	δ_{peak} (mm)	σ_n (MPa)	τ_{peak} (MPa)	$\delta_{v,peak}$ (mm)	$\delta_{h,\delta_v=0}$ (mm)	$\delta_{v,min}$ (mm)	Post-peak shear displacements (mm)				Post-peak shear stress (MPa)				Post-peak dilation displacement (mm)			
										1	2	3	4	1	2	3	4	1	2	3	4
Bandis et al. (1981); Figs. 9–21; Bedding plane in course grained sandstone; $\phi_b=32°$																					
228	0.05	18.5	2	0.7	24.5	0.059	0.02	0.34	0.00	2	3	4	–	0.050	0.045	0.042	–	0.26	0.5	0.64	–
229	0.10	15.0	1.361	0.8	24.5	0.039	0.03	0.59	0.00	2	3	4	6	0.041	0.038	0.037	0.035	0.3	0.47	0.61	–
230	0.20	12.2	0.927	1.1	24.5	0.031	0.12	0.34	0.00	2	3	4	6	0.031	0.030	0.030	0.030	0.27	0.43	0.54	0.66
231	0.40	9.9	0.631	1.9	24.5	0.025	0.11	0.56	0.00	2	3	4	6	0.026	0.026	0.026	0.026	0.17	0.3	0.44	0.56
Huang et al. (2002); Figs. 3–8; Replica; $\phi_b=35°$																					
232	0.10	8.7	8.5	0.6	100	0.103	0.05	0	0.00	2	3	4	–	0.112	0.118	0.121	–	0.44	0.72	1.04	–
233	0.10	8.7	8.5	1.16	300	0.413	0	1.05	−0.04	2	3	4	–	0.434	0.430	0.423	–	0.1	0.25	0.45	–
234	0.10	8.7	8.5	0.99	500	0.523	0	1.57	−0.03	2	3	4	–	0.500	0.478	0.463	–	0.02	0.08	0.14	–
235	0.10	8.7	8.5	1.49	1000	0.929	−0.03	1.57	−0.03	2	3	4	–	0.898	0.861	0.851	–	0	0	−0.01	–
236	0.10	8.7	8.5	2.48	1500	1.278	−0.07		−0.06	3	4	–	–	1.257	1.192	–	–	−0.06	−0.06	–	–
237	0.10	15.8	8.5	0.84	100	0.191	0.07	0.54	0.00	2	3	–	–	0.210	0.210	–	–	0.75	1.3	–	–
238	0.10	15.8	8.5	1.16	300	0.606	0.04	0.39	−0.04	2	3	4	–	0.570	0.520	0.500	–	0.2	0.32	0.44	–
239	0.10	15.8	8.5	2	500	0.785	0.03	1.42	0.00	3	4	–	–	0.760	0.730	–	–	0.13	0.27	–	–
240	0.10	15.8	8.5	1.84	1000	1.067	0.01	1.42	0.00	2	3	4	–	1.040	0.920	0.820	–	0.01	0.01	0	–
241	0.10	15.8	8.5	2.5	1500	1.530	−0.02	–	−0.03	3	4	–	–	1.480	1.390	–	–	−0.03	−0.03	–	–
Wibowo (1994) and Wibowo et al. (1993); G2; Replica; $\phi_b=32°$																					
242	0.14	14.6	27.6	1.28	276	0.381	0.128	–	–	13	–	–	–	0.236	–	–	–	–	–	–	–
243	0.14	14.6	27.6	1.94	1380	1.479	0.1552	–	–	13	–	–	–	1.139	–	–	–	–	–	–	–
244	0.14	14.6	27.6	2.29	2755	30.401	0.2748	–	–	13	–	–	–	2.493	–	–	–	–	–	–	–
245	0.14	14.6	27.6	3.95	5516	5.488	0.0395	–	–	13	–	–	–	4.171	–	–	–	–	–	–	–
Wibowo (1994) and Wibowo et al. (1993); F2; Replica; $\phi_b=32°$																					
246	0.10	9.9	27.6	0.34	283.3	0.306	0.0544	–	–	13	–	–	–	0.104	–	–	–	–	–	–	–
247	0.10	9.9	27.6	0.55	1375	1.422	0.066	–	–	13	–	–	–	0.768	–	–	–	–	–	–	–
248	0.10	9.9	27.6	1.07	2749	2.417	0.0642	–	–	13	–	–	–	1.547	–	–	–	–	–	–	–
249	0.10	9.9	27.6	1.08	5509	4.482	0.108	–	–	13	–	–	–	3.316	–	–	–	–	–	–	–
Wibowo (1994) and Wibowo et al. (1993); F3; Replica; $\phi_b=32°$																					
250	0.11	9.8	27.6	0.10	272.7	0.367	0.038	–	–	15	–	–	–	0.136	–	–	–	–	–	–	–
251	0.11	9.8	27.6	0.67	1380	1.297	0.067	–	–	15	–	–	–	0.843	–	–	–	–	–	–	–
252	0.11	9.8	27.6	0.78	2760	3.232	0.117	–	–	15	–	–	–	2.23	–	–	–	–	–	–	–
253	0.11	9.8	27.6	2.08	5512	4.233	0.208	–	–	15	–	–	–	3.386	–	–	–	–	–	–	–
Wibowo (1994) and Wibowo et al. (1993); F4; Replica; $\phi_b=32°$																					
254	0.11	15.5	27.6	0.42	275.6	0.464	0.0504	–	–	15	–	–	–	0.223	–	–	–	–	–	–	–
255	0.11	15.5	27.6	1.34	1378	1.532	0.2144	–	–	15	–	–	–	0.812	–	–	–	–	–	–	–
256	0.11	15.5	27.6	1.56	2756	2.845	0.1872	–	–	15	–	–	–	1.821	–	–	–	–	–	–	–
257	0.11	15.5	27.6	1.90	4134	4.370	0.171	–	–	15	–	–	–	3.015	–	–	–	–	–	–	–
258	0.11	15.5	27.6	2.09	5512	55.263	0.1881	–	–	15	–	–	–	3.813	–	–	–	–	–	–	–

Table 9
Monotonic Direct Shear Test (MDST) database; Part 2: without post-peak behavior information.

No	L (m)	JRC	JCS (MPa)	δ_{peak} (mm)	σ_n (MPa)	τ_{peak} (MPa)
Grasselli and Egger (2003); Sandstone; $\phi_b = 37°$						
259	0.14	27.2	10	0.65	1020	2.088
260	0.14	16.3	10	0.67	4130	3.886
261	0.14	15.0	10	0.67	2090	2.257
Grasselli and Egger (2003); Gneiss; $\phi_b = 36°$						
262	0.14	12.9	184	0.48	1900	3.400
263	0.14	7.4	184	0.2	3520	4.000
264	0.14	6.7	184	0.31	3570	3.900
265	0.14	8.6	184	0.35	3520	4.300
266	0.14	1.8	184	0.31	4080	3.300
267	0.14	9.4	184	0.35	2600	3.500
268	0.14	13.0	87	0.28	870	1.707
Grasselli and Egger (2003); Limestone; $\phi_b = 36°$						
269	0.14	20.5	25	0.5	1070	2.200
270	0.14	19.7	25	0.52	1070	2.100
271	0.14	24.1	25	0.53	3720	5.500
272	0.14	25.7	25	0.31	2450	4.600
273	0.14	24.4	25	0.24	3110	5.000
274	0.14	20.2	25	0.37	1020	2.100
275	0.14	23.9	25	0.74	3110	4.900
Grasselli and Egger (2003); Granite; $\phi_b = 34°$						
276	0.14	18.1	173	0.38	2300	5.700
277	0.14	17.9	173	0.65	2300	5.600
278	0.14	16.6	173	0.45	2190	4.800
279	0.14	14.2	173	0.62	1120	2.400
280	0.14	15.9	173	0.38	1120	2.900
281	0.14	15.6	173	0.23	1120	2.800
282	0.14	16.2	173	0.56	1120	3.000
283	0.14	3.3	184	0.3	2650	2.400
Grasselli and Egger (2003); Marble; $\phi_b = 37°$						
284	0.14	11.7	87	0.27	1.73	2.655
285	0.14	16.6	87	0.5	0.87	2.417
286	0.14	13.5	87	0.88	3.78	5.477
287	0.14	9.2	87	0.29	2.6	3.214
288	0.14	9.0	87	0.44	2.6	3.179
289	0.14	14.8	87	0.44	3.78	5.856
290	0.14	15.0	87	0.39	383	5.976
291	0.14	14.3	87	0.42	2.6	4.293
292	0.14	13.7	87	0.27	0.87	1.816
293	0.14	16.3	87	0.55	1.79	3.752
Aydan et al. (1990); Plaster; $\phi_b = 36°$						
294	0.16	11.0	0.92	2.45	0.04	0.023
295	0.16	13.0	0.92	1.20	0.03	0.0307
296	0.16	13.0	0.92	2.00	0.07	0.131
297	0.16	11.0	0.92	1.80	0.02	0.046
Bandis et al. (1981); Figs. 9–21; Bedding plane in course grained sandstone; $\phi_b = 32°$						
298	0.05	18.5	2	0.7	24.5	0.059
299	0.10	15.0	1.419	0.8	24.5	0.040
300	0.20	12.2	1.007	1.1	24.5	0.031
301	0.40	9.9	0.715	1.7	24.5	0.026
Bandis et al. (1981); Figs. 9–21; Bedding plane in limestone; $\phi_b = 32°$						
302	0.06	17.5	2000	0.75	0.0245	0.055
303	0.12	14.2	1404.4	0.91	0.0245	0.041
304	0.18	12.6	1142.1	1.87	0.0245	0.046
305	0.36	10.2	802	2.17	0.0245	0.042
Huang et al. (1996); Figs. 3–8; Replica; $\phi_b = 35°$						
306	0.10	0.0	8.5	0.33	300	0.184
307	0.10	0.0	8.5	0.65	500	0.308
308	0.10	0.0	8.5	1.96	1000	0.777
308	0.10	0.0	8.5	2.71	1500	303
Yoshinaka and Yamabe (1986); Diamond Sawed; Welded-tuff; $\phi_b = 39°$						
310	0.14	0.0	11.2	1.25	0.490	0.229
311	0.14	0.0	11.2	1.45	0.980	1.14
312	0.14	0.0	11.2	1.55	1.460	0.77
313	0.14	0.0	11.2	1.55	1.950	0.57

(continued on next page)

Table 9 (*continued*)

No	*L* (m)	*JRC*	*JCS* (MPa)	δ_{peak} (mm)	σ_n (MPa)	τ_{peak} (MPa)
314	0.14	0.0	11.2	1.80	2.930	3.8
Yoshinaka and Yamabe (1986); Sand Blasted; Welded-tuff; $\phi_b=33°$						
315	0.13	0.0	11.2	0.25	0.48	0.233
316	0.13	0.0	11.2	1.10	0.91	1.23
317	0.13	0.0	11.2	0.75	1.19	0.94
318	0.13	0.0	11.2	0.90	1.80	0.62
319	0.13	0.0	11.2	1.20	2.60	0.43
320	0.13	0.0	11.2	1.30	3.05	3.7
Van Sint Jan (1990); Plaster; $\phi_b=48°$						
321	0.15	20.0	14.7	1.50	0.5	2.94
322	0.15	20.0	14.7	1.70	1	1.47
323	0.15	20.0	14.7	1.70	2	7.35
324	0.15	20.0	14.7	2.20	4	3.675
325	0.15	20.0	14.7	2.40	6	2.45
Desai and Fishman (1991); Figs. 11–18; Concrete replica; $\phi_b=30°$						
326	0.30	0.0	30	0.05	0.035	0.021
327	0.30	0.0	30	0.1	0.138	0.080
328	0.30	0.0	30	0.75	0.345	0.199
329	0.30	2.0	30	1.2	0.035	0.024
330	0.30	3.0	30	1.1	0.035	0.025
331	0.30	2.0	30	1.25	0.069	0.047
332	0.30	3.0	30	1.2	0.069	0.050
333	0.30	2.0	30	1.6	0.138	0.093
334	0.30	3.0	30	2.5	0.138	0.100
335	0.30	2.0	30	1.8	0.345	0.233
336	0.30	3.0	30	4	0.345	0.251
Bandis et al. (1981); Figs. 9–21; Tension joints in siltstone; $\phi_b=32°$						
337	0.06	17.5	2	1.13	0.0245	0.045
338	0.12	14.2	1.4	1.74	0.0245	0.033
339	0.18	12.6	1.14	2.13	0.0245	0.035
340	0.36	10.2	0.80	3.5	0.0245	0.026
341	0.06	17.5	2	1.13	0.0245	0.055
342	0.12	14.2	1.48	1.22	0.0245	0.039
343	0.18	12.6	1.24	1.45	0.0245	0.034
344	0.36	10.2	0.917	2.17	0.0245	0.028
Bandis et al. (1981); Figs. 9–21; Bedding plane in limestone; $\phi_b=32°$						
345	0.06	17.5	2	0.6	0.0245	0.062
346	0.12	14.2	1.464	0.7	0.0245	0.050
347	0.18	12.6	1.220	0.9	0.0245	0.034
348	0.36	10.2	0.893	1.8	0.0245	0.027
Bandis et al. (1981); Figs. 9–21; Bedding plane in slightly metamorphosed fine grained sandstone; $\phi_b=32°$						
349	0.06	17.5	2	1.13	0.0245	0.043
350	0.12	14.2	1.511	1.37	0.0245	0.041
351	0.18	12.6	1.282	1.67	0.0245	0.032
352	0.36	10.2	0.968	2.91	0.0245	0.030
353	0.06	17.5	2	0.9	0.0245	0.037
354	0.12	14.2	1.558	1.05	0.0245	0.032
355	0.18	12.6	1.347	1.15	0.0245	0.028
356	0.36	10.2	1.049	1.4	0.0245	0.024
357	0.06	17.5	2	0.75	0.0245	0.028
358	0.12	14.2	1.694	1.15	0.0245	0.026
359	0.18	12.6	1.537	1.1	0.0245	0.024
360	0.36	10.2	1.301	1.2	0.0245	0.022
Grasselli and Egger (2003); Serpentines; $\phi_b=39°$						
361	0.14	17.0	74	0.4	1940	4.334
362	0.14	20.9	74	0.5	970	4.702

Along the X_i axis, $(JRC_i)_1=(JRC_i)_2=0, (L_i)_1=(L_i)_2$ and $(\sigma_i)_1=(\sigma_i)_2$. Therefore, the peak shear displacements along the X_i direction are equal ($(\delta_i)_1=(\delta_i)_2$) and Eq. (40) can be simplified as:

$$\frac{\delta_1}{\delta_2}=\frac{\cos(i_1)}{\cos(i_2)} \quad (41)$$

Because $0<i_1<i_2<90$, we have $\cos i_1>\cos i_2$. Thus, it can be concluded that the peak shear displacement decreases with the increasing dilation angle (or *JRC*):

$$\delta_1>\delta_2 \quad (42)$$

Appendix C

Eq. (15):

The goal is to find an empirical equation for the peak shear displacement of sawed fracture, δ_i, and then obtain the peak shear displacement of rough fractures, δ_{peak}, using Eq. (34).

By performing a dimensional analysis (Szirtes, 1997), the following dimensionless parameters were found:

$$\pi_7=\frac{\delta_i}{L_i}, \pi_8=\frac{L_i}{L_0}, \pi_9=\frac{\sigma_i}{JCS},$$

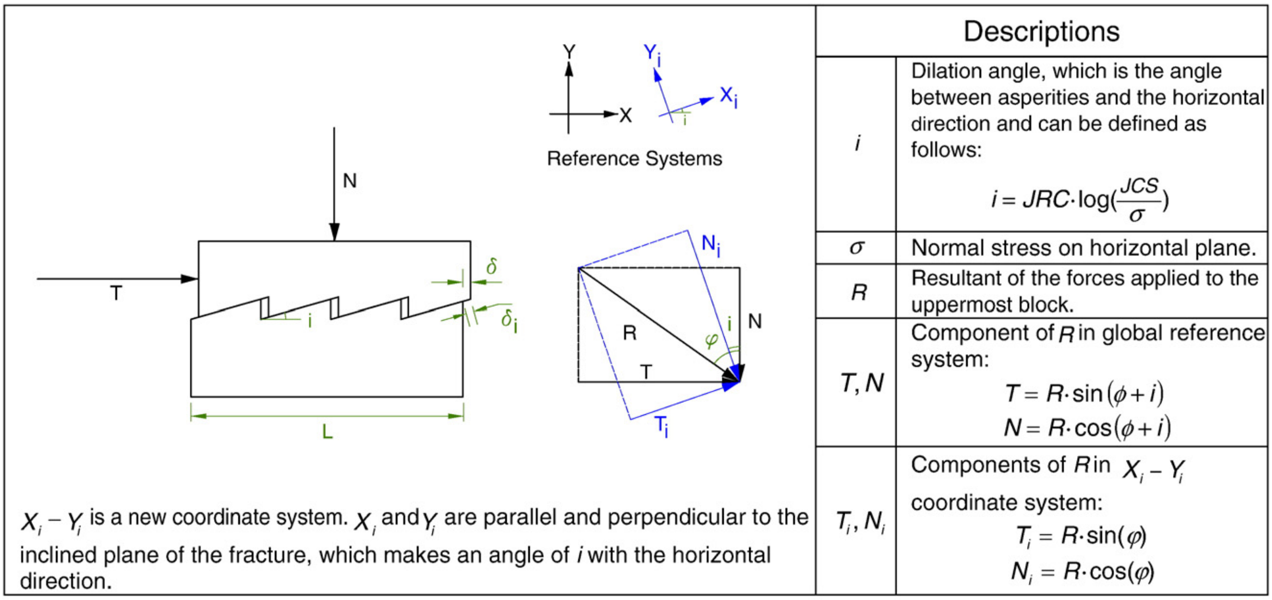

Fig. 12. Diagram of forces applied at failure in shearing rough fracture.

where L_0 is length of lab specimen (0.1 m). Based on Eqs. (32)–(34), they can be written as follows:

$$\pi_7 = \frac{\delta}{L},\ \pi_8 = \frac{L}{L_0 \cdot \cos(i)},\ \pi_9 = \frac{\sigma_n}{JCS} \cdot \frac{\cos(\phi) \cdot \cos(i)}{\cos(\phi + i)}$$

All 366 direct shear test records available in the MDST database were used to calculate the three above mentioned dimensionless parameters and perform correlation analyses to find π_7 as a function of π_8 and π_9:

$$\pi_7 = f(\pi_8, \pi_9) \tag{43}$$

Since the power function is a convenient form to be used in calculations, it was assumed that function f in Eq. (43) is a power function. Linear multivariable regression analysis was performed. With R square of 0.38 and standard error of estimate of 0.68, the following equation for the peak shear displacement of sawed fractures was obtained (adopting SI units):

$$\delta_i = 0.0077 L_i^{0.45} \left(\frac{\sigma_i}{JCS}\right)^{0.34} \tag{44}$$

Substituting Eqs. (32)–(34) into Eq. (44), the peak shear displacement of the rough fractures can be obtained using the following equation:

$$\delta_{peak} = 0.0077 L^{0.45} \left(\frac{\sigma_n}{JCS}\right)^{0.34} \cos(i) \cdot \left[\frac{1}{(\cos i)^{0.45}} \left(\frac{\cos\phi \cdot \cos i}{\cos(\phi + i)}\right)^{0.34}\right] \tag{45}$$

The base friction angle, ϕ, ranges between 25° to 35° for the majority of unweathered rock surfaces (Barton, 1971; Coulson, 1972; Hutchinson, 1972; Krsmanovic, 1967; Patton, 1966b; Ripley and Lee, 1962; Wallace et al., 1970). Furthermore, in view of the safety requirement of rock engineering structures, the value of $\phi + i$ is limited to 70°. Therefore, the last part of the right hand side of Eq. (45), $\left[\frac{1}{(\cos i)^{0.45}} \left(\frac{\cos\phi \cdot \cos i}{\cos(\phi + i)}\right)^{0.34}\right]$, can range between 1 and 1.45. Thus, for simplicity it can be eliminated from the equation. Consequently, the following empirical relation can be used for the peak shear displacement of rock fractures:

$$\delta_{peak} = 0.0077 \cdot L^{0.45} \cdot \left(\frac{\sigma_n}{JCS}\right)^{0.34} \cdot \cos\left(JRC \cdot \log\left(\frac{JCS}{\sigma_n}\right)\right) \tag{46}$$

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